

Relations

Chapter 9

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Chapter Summary

9.1 Relations and Their Properties

9.2 n -ary Relations and Their Applications

9.3 Representing Relations

9.4 Closures of Relations

9.5 Equivalence Relations

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Relations and Their Properties

Section 9.1

9.1.1 Introduction

To express a relationship between elements of two sets is to use **ordered pairs** made up of two related elements. For this, sets of ordered pairs are called **binary relations**.

DEFINITION 1: Let A and B be sets. A **binary relation** (R) from A to B is a subset of $A \times B$.

$$R \subseteq A \times B.$$

In other words, a binary relation from A to B is **a set R of ordered pairs**, where the **first** element of each ordered pair comes from A and the **second** element comes from B.

□ **aRb** to denote that $(a, b) \in R$

□ **$a \not R b$** to denote that $(a, b) \notin R$.

Binary relations

Binary relations represent relationships between the elements of **two sets**.

We will introduce **n-ary relations**, which express relationships among elements of more than two sets, later in this chapter.

We will **omit the word binary** when there is no danger of confusion.

Binary Relations

Let A be the set of **students** in school, and let B be the set of **courses**.
Let R be the relation that consists of those **pairs** (a, b) , where a is a student enrolled in course b .

For instance, if Jason and Deborah are enrolled in CS518,
the pairs $(\text{Jason}, \text{CS518})$ and $(\text{Deborah}, \text{CS518})$ belong to R .

If Jason is also enrolled in CS510,
then the pair $(\text{Jason}, \text{CS510})$ is also in R .

However, if Deborah Sherman is not enrolled in CS510,
then the pair $(\text{Deborah}, \text{CS510})$ is not in R .

Note that if a student is not currently enrolled in any courses there will be **no pairs in R** that have this student as the **first element**.

Similarly, if a course is not currently being offered there will be no pairs in R that have this course as their **second element**.

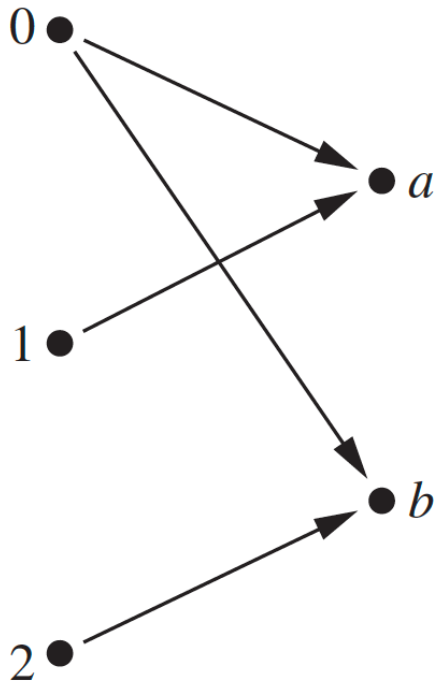
Binary Relations

Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$.

$\{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B .

This means, for instance, that $0Ra$, but that $\neg 1Rb$.

Relations can be represented graphically or using a table.



R	a	b
0	×	×
1	×	
2		×

using arrows to represent ordered pairs.

9.1.2 Functions as Relations

Recall that a **function** f from a set A to a set B (as defined in Section 2.3) assigns **exactly one element** of B to each element of A .

The graph of f is the set of ordered pairs (a, b) such that $b = f(a)$.

Because the graph of f is a subset of $A \times B$, it is a **relation** from A to B .

$$(a, f(a)) \text{ for } a \in A$$

Moreover, the graph of a function has the **property** that every element of A is the first element of exactly one ordered pair of the graph.

Relations are a **generalization** of graphs of functions; they can be used to express a much wider class of relationships between sets.

9.1.3 Relations on a Set

Relations from a set A to itself are of special interest.

DEFINITION 2: A relation on a set A is a relation from A to A .

In other words, a relation on a set A is a subset of $A \times A$.

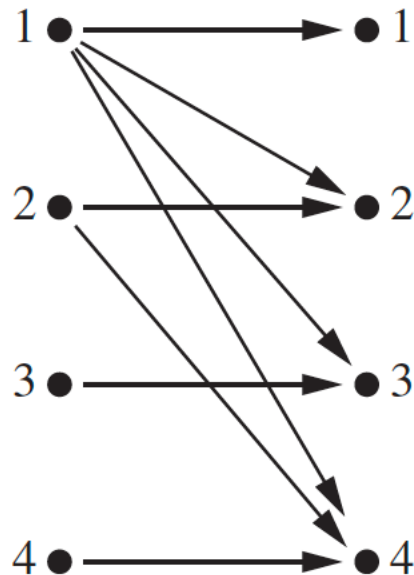
EXAMPLE: $A = \{a, b, c\}$.

Then $R = \{(a, a), (a, b), (a, c)\}$ is a relation on A .

Binary Relations on a Set₁

Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$.



R	1	2	3	4
1	×	×	×	×
2		×		×
3			×	
4				×

Binary Relations on a Set₂

How many relations are there on a set A with n elements?

Solution:

A relation on a set A is a subset of $A \times A$.

Because $A \times A$ has n^2 elements when A has n elements, and a set with m elements has 2^m subsets (the cardinality of the power set), there are 2^{n^2} subsets of $A \times A$.

Thus, there are 2^{n^2} relations on a set with n elements.

For EXAMPLE, there are $2^{3^2} = 2^9 = 512$ relations on the set $\{a, b, c\}$.

Binary Relations on a Set₃

EXAMPLE: Consider these relations on the set of integers:

$$R_1 = \{(a, b) | a \leq b\},$$

$$R_2 = \{(a, b) | a > b\},$$

$$R_3 = \{(a, b) | a = b \text{ or } a = -b\},$$

$$R_4 = \{(a, b) | a = b\},$$

$$R_5 = \{(a, b) | a = b + 1\},$$

$$R_6 = \{(a, b) | a + b \leq 3\}.$$

Which of these relations contain each of the pairs (1,1), (1, 2), (2, 1), (1, -1), and (2, 2)?

(1,1) is in R_1, R_3, R_4 , and R_6

(1,2) is in R_1 and R_6

(2,1) is in R_2, R_5 , and R_6

(1, -1) is in R_2, R_3 , and R_6

(2,2) is in R_1, R_3 , and R_4 .

Note that these relations are on an infinite set and each of these relations is an infinite set.

9.1.4 Properties of Relations

Reflexive

Symmetric, Antisymmetric

Transitive

Reflexive Relations

DEFINITION 3: A relation R on a set A is called **reflexive** if $(a, a) \in R$ for every element $a \in A$.

R is **reflexive** iff $(a, a) \in R$ for every element $a \in A$.

Written symbolically, R is **reflexive** if and only if

$$\forall x (x \in U \rightarrow (x, x) \in R)$$

$$\forall a ((a, a) \in R)$$

If $A = \emptyset$ then the empty relation is reflexive vacuously.
That is the empty relation on an empty set is reflexive!

Reflexive Relations

Consider the following relations on $\{1, 2, 3, 4\}$:

$$R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$$

$$R2 = \{(1, 1), (1, 2), (2, 1)\},$$

$$R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$$

$$R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$$

$$R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$$

$$R6 = \{(3, 4)\}.$$

Which of these relations are reflexive?

$$\forall x (x \in U \rightarrow (x, x) \in R)$$

Reflexive Relations

Consider the following relations on $\{1, 2, 3, 4\}$:

$$R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}, \quad \times$$

$$R2 = \{(1, 1), (1, 2), (2, 1)\}, \quad \times$$

$$R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}, \quad \checkmark$$

$$R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}, \quad \times$$

$$R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}, \quad \checkmark$$

$$R6 = \{(3, 4)\}. \quad \times$$

Which of these relations are reflexive?

$$\forall x (x \in U \rightarrow (x, x) \in R)$$

The relations **R3 and R5 are reflexive** because they both contain all pairs of the form (a, a) , namely, $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$.

In particular, **R1, R2, R4, and R6 are not reflexive** because $(3, 3)$ is not in any of these relations.

Reflexive Relations

Which of these relations are reflexive?

✓ $R_1 = \{(a, b) | a \leq b\},$

✗ $R_2 = \{(a, b) | a > b\},$ $3 \not> 3$

✓ $R_3 = \{(a, b) | a = b \text{ or } a = -b\},$

✓ $R_4 = \{(a, b) | a = b\},$

✗ $R_5 = \{(a, b) | a = b + 1\},$ $3 \neq 3 + 1$

✗ $R_6 = \{(a, b) | a + b \leq 3\}.$ $3 + 3 \not\leq 3$

$$\forall x (x \in U \rightarrow (x, x) \in R)$$

Reflexive Relations

Is the “divides” relation on the set of positive integers reflexive?

✓ Because $a \mid a$ whenever a is a positive integer, the “divides” relation is reflexive.

Is the “divides” relation on the set of all integers reflexive?

✗ The relation is not reflexive because by DEFINITION 0 does not divide 0.)

$$\forall x (x \in U \rightarrow (x, x) \in R)$$

Symmetric and Antisymmetric Relations

DEFINITION 4

A relation R on a set A is called **symmetric** if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$.

$$\forall x \forall y ((x, y) \in R \rightarrow (y, x) \in R)$$

A relation R on a set A such that for all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called **antisymmetric**.

$$\forall x \forall y ((x, y) \in R \wedge (y, x) \in R \rightarrow x = y)$$

Symmetric and Antisymmetric Relations

symmetric

$$\forall x \forall y ((x, y) \in R \rightarrow (y, x) \in R)$$

antisymmetric

$$\forall x \forall y ((x, y) \in R \wedge (y, x) \in R \rightarrow x = y)$$

The terms symmetric and antisymmetric are **not opposites**, because a relation can have **both** of these properties or may **lack both** of them.

□ $\{(a, a), (b, b), (c, c)\}$ on $\{a, b, c\}$

A relation **cannot be both** symmetric and antisymmetric if it contains some pair of the form **(a, b) in which $a \neq b$** .

□ $\{(a, b), (b, a), (a, c)\}$ on $\{a, b, c\}$

Symmetric and Antisymmetric Relations

$$\forall x \forall y ((x, y) \in R \rightarrow (y, x) \in R)$$

Which of these relations are **symmetric**?

Consider the following relations on $\{1, 2, 3, 4\}$:

$R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$, 

$R2 = \{(1, 1), (1, 2), (2, 1)\}$, 

$R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$, 

$R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$, 

$R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$, 

$R6 = \{(3, 4)\}$, 

Symmetric and Antisymmetric Relations

$$\forall x \forall y ((x, y) \in R \wedge (y, x) \in R \rightarrow x = y)$$

Which of these relations are **antisymmetric**?

Consider the following relations on $\{1, 2, 3, 4\}$:

$R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$, 

$R2 = \{(1, 1), (1, 2), (2, 1)\}$, 

$R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$, 

$R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$, 

$R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$, 

$R6 = \{(3, 4)\}$, 

Symmetric and Antisymmetric Relations

$$\forall x \forall y ((x, y) \in R \rightarrow (y, x) \in R)$$

Which of these relations are **symmetric**?

✗ $R_1 = \{(a, b) | a \leq b\}, \quad 3 \leq 4, \text{ but } 4 \not\leq 3$

✗ $R_2 = \{(a, b) | a > b\}, \quad 4 > 3, \text{ but } 3 \not> 4$

✓ $R_3 = \{(a, b) | a = b \text{ or } a = -b\},$

✓ $R_4 = \{(a, b) | a = b\},$

✗ $R_5 = \{(a, b) | a = b + 1\}, \quad 3 = 2 + 1, \text{ but } 2 \neq 3 + 1$

✓ $R_6 = \{(a, b) | a + b \leq 3\}.$

Symmetric and Antisymmetric Relations

$$\forall x \forall y ((x, y) \in R \wedge (y, x) \in R \rightarrow x = y)$$

Which of these relations are **antisymmetric**?

✓ $R_1 = \{(a, b) | a \leq b\},$

✓ $R_2 = \{(a, b) | a > b\},$

✗ $R_3 = \{(a, b) | a = b \text{ or } a = -b\}, (1, -1) \text{ and } (-1, 1)$

✓ $R_4 = \{(a, b) | a = b\},$

✓ $R_5 = \{(a, b) | a = b + 1\},$

✗ $R_6 = \{(a, b) | a + b \leq 3\}. (1, 2) \text{ and } (2, 1)$

Symmetric and Antisymmetric Relations

Is the “divides” relation on the set of positive integers **symmetric**? Is it **antisymmetric**?

$$\forall x \forall y ((x, y) \in R \rightarrow (y, x) \in R) \quad \forall x \forall y ((x, y) \in R \wedge (y, x) \in R \rightarrow x = y)$$

This relation is **not symmetric**.

$1|2$, but $2 \nmid 1$.

However, it is **antisymmetric**.

if $a|b$ and $b|a$, then $a = b$

Transitive Relations

Let R be the relation consisting of all pairs (x, y) of students at your school, where x has taken more credits than y .

If (x, y) and (y, z) , then (x, z) .

What we have shown is that R has the **transitive** property, which is defined as follows.

DEFINITION 5: A relation R on a set A is called **transitive** if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

$$\forall x \forall y \forall z ((x, y) \in R \wedge (y, z) \in R \rightarrow (x, z) \in R)$$

Transitive Relations

$$\forall x \forall y \forall z [(x, y) \in R \wedge (y, z) \in R \rightarrow (x, z) \in R]$$

Which of these relations are **transitive**?

✓ $R_1 = \{(a, b) | a \leq b\},$

✓ $R_2 = \{(a, b) | a > b\},$

✓ $R_3 = \{(a, b) | a = b \text{ or } a = -b\},$

✓ $R_4 = \{(a, b) | a = b\},$

✗ $R_5 = \{(a, b) | a = b + 1\},$ (4,3), (3,2), but (4,2) $\notin R$

✗ $R_6 = \{(a, b) | a + b \leq 3\}.$ (2,1), (1,2), but (2,2) $\notin R$

Transitive Relations

Which of these relations are **transitive**?

Consider the following relations on $\{1, 2, 3, 4\}$:

$$R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}, \quad \times$$

$$R2 = \{(1, 1), (1, 2), (2, 1)\}, \quad \times$$

$$R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}, \quad \times$$

$$R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}, \quad \checkmark$$

$$R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}, \quad \checkmark$$

$$R6 = \{(3, 4)\}. \quad \checkmark$$

How many Reflexive & Symmetric & Antisymmetric Relations

How many relations are there on a set with n elements that are

reflexive?

$$2^{n(n-1)}$$

symmetric?

$$2^{n(n+1)/2}$$

antisymmetric?

$$2^n 3^{n(n-1)/2}$$

irreflexive?

$$2^{n(n-1)}$$

asymmetric?

$$3^{n(n-1)/2}$$

reflexive and symmetric?

$$2^{n(n-1)/2}$$

neither reflexive nor irreflexive?

$$2^{n^2} - 2 \cdot 2^{n(n-1)}$$

How many Reflexive & Symmetric & Antisymmetric & Transitive Relations

How many **reflexive** relations are there on a set with n elements?

- 1) A relation with n^2 ordered pairs in $A \times A$ is in R .
- 2) However, if R is reflexive, each of the n ordered pairs (a, a) for $a \in A$ must be in R .
- 3) Each of the **other** $n(n - 1)$ ordered pairs of the form (a, b) , where $a \neq b$, may or may not be in R .
- 4) Hence, by the product rule for counting, there are $2^{n(n-1)}$ reflexive relations.

$$1(1)$$

$$2(4)$$

$$3(64) \dots\dots$$

9.1.5 Combining Relations

Given two relations R_1 and R_2 , we can combine them using basic set operations to form new relations such as $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, and $R_2 - R_1$.

EXAMPLE: Let $A = \{1,2,3\}$ and $B = \{1,2,3,4\}$.

$R_1 = \{(1,1),(2,2),(3,3)\}$ and $R_2 = \{(1,1),(1,2),(1,3),(1,4)\}$ can be combined using basic set operations to form new relations:

$$R_1 \cup R_2 = \{(1,1),(1,2),(1,3),(1,4),(2,2),(3,3)\}$$

$$R_1 \cap R_2 = \{(1,1)\}$$

$$R_2 - R_1 = \{(1,2),(1,3),(1,4)\}$$

$$R_1 - R_2 = \{(2,2),(3,3)\}$$

9.1.5 Combining Relations

EXAMPLE: Let R_1 be the **less than** relation on the set of real numbers and let R_2 be the **greater than** relation on the set of real numbers, that is, $R_1 = \{(x, y) \mid x < y\}$ and $R_2 = \{(x, y) \mid x > y\}$.

$$\square R_1 \cup R_2 = \{(x, y) \mid x \neq y\}.$$

$$\square R_1 \cap R_2 = \emptyset$$

$$\square R_1 - R_2 = R_1$$

$$\square R_2 - R_1 = R_2$$

$$\square R_1 \oplus R_2 = R_1 \cup R_2 - R_1 \cap R_2 = \{(x, y) \mid x \neq y\}.$$

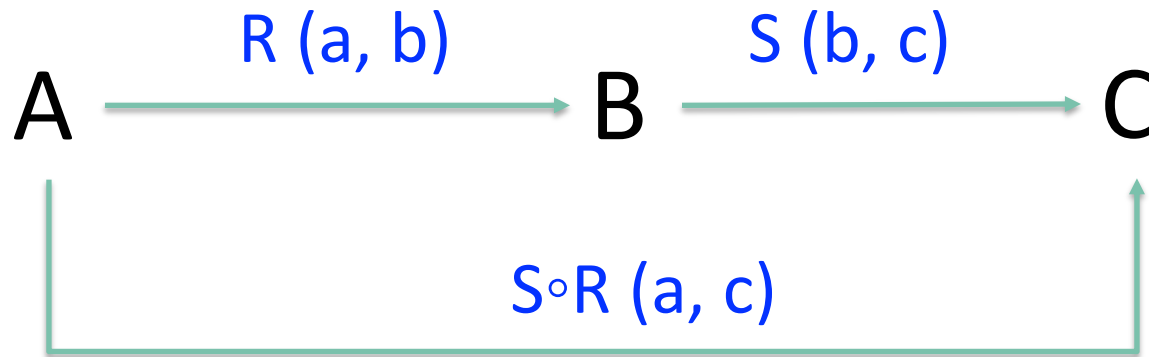
Composition

DEFINITION 6 :

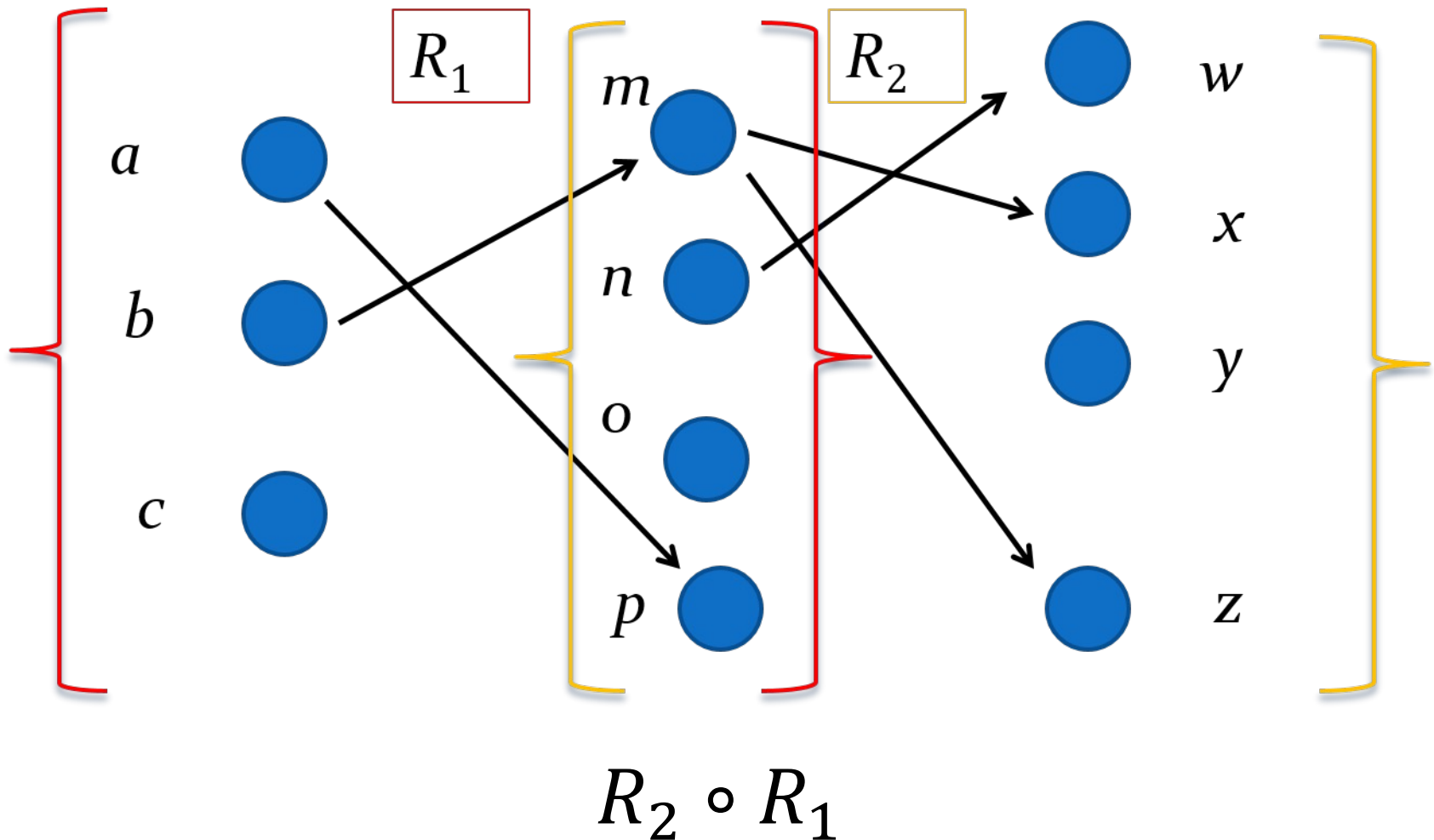
Let R be a relation from a set A to a set B and S a relation from B to a set C .

The composite of R and S is the relation consisting of ordered pairs (a, c) , where $a \in A$, $c \in C$, and for which there exists an element $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$.

We denote the **composite** of R and S by $S \circ R$.



Representing the Composition of Relations



Powers of a Relation

DEFINITION 7: Let R be a binary relation on A . Then the powers R^n of the relation R can be defined inductively by:

- Basis Step: $R^1 = R$
- Inductive Step: $R^{n+1} = R^n \circ R$

The powers of a transitive relation are subsets of the relation. This is established by the following THEOREM:

THEOREM 1: The relation R on a set A is **transitive** iff $R^n \subseteq R$ for $n = 1, 2, 3, \dots$

(see the text for a proof via mathematical induction)

Powers of a Relation

EXAMPLE: Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$. Find the powers R^n , $n = 2, 3, 4, \dots$

□ $R^2 = R \circ R = \{(1, 1), (2, 1), (3, 1), (4, 2)\}$.

□ $R^3 = R^2 \circ R = \{(1, 1), (2, 1), (3, 1), (4, 1)\}$.

□ R^4 is the same as R^3 , so $R^4 = \{(1, 1), (2, 1), (3, 1), (4, 1)\}$.

□ It also follows that $R^n = R^3$ for $n = 4, 5, 6, 7, \dots$

Section Summary

Relations and Functions

Properties of Relations

- Reflexive Relations
- Symmetric and Antisymmetric Relations
- Transitive Relations

Combining Relations

Homework after class

9.1 Relations and Their Properties

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n-ary Relations and Their Applications

Section 9.2

9.2.1 Introduction

We will study relationships among elements from **more than two sets**.

- ❑ There is a relationship involving the name of a **student**, the student's major, and the student's grade point average.
- ❑ There is a relationship involving the **airline**, flight number, starting point, destination, departure time, and arrival time of a flight.
- ❑ An EXAMPLE of such a relationship in **mathematics** involves three integers, where the first integer is larger than the second integer, which is larger than the third.
- ❑ The betweenness relationship involving **points on a line**, such that three points are related when the second point is between the first and the third.

9.2.1 Introduction

Relationships among elements of **more than two sets** often arise.

These relationships are called **n-ary relations**.

These relations are used to represent **computer databases**. These representations **help us answer queries** about the information stored in databases, such as:

- ☐ Which **flights** land at O'Hare Airport between 3 A.M. and 4 A.M.?
- ☐ Which **students** at your school are sophomores majoring in mathematics or computer science and have greater than a 3.0 average?
- ☐ Which **employees** of a company have worked for the company less than 5 years and make more than \$50,000?

9.2.2 n-ary Relations

DEFINITION 1: Let $A_1, A_2, \dots, A_n$ be sets. An **n-ary relation** on these sets is a subset of $A_1 \times A_2 \times \dots \times A_n$. The sets $A_1, A_2, \dots, A_n$ are called the **domains** of the relation, and n is called its **degree**.

EXAMPLE: Let R be the relation on $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$ consisting of triples (a, b, c) , where a, b , and c are integers with $a < b < c$.

- ❑ Then $(1, 2, 3) \in R$, but $(2, 4, 3) \notin R$.
- ❑ The degree of this relation is 3.
- ❑ Its domains are all equal to the set of natural numbers.

9.2.2 n-ary Relations

EXAMPLE: Let R be the relation on $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ consisting of all triples of integers (a, b, c) in which a , b , and c form an arithmetic progression.

- That is, $(a, b, c) \in R$ if and only if there is an integer k such that $b = a + k$ and $c = a + 2k$, or equivalently, such that $b - a = k$ and $c - b = k$.
- Note that $(1, 3, 5) \in R$, but $(2, 5, 9) \notin R$.
- This relation has degree 3.
- Its domains are all equal to the set of integers.

9.2.2 n-ary Relations

EXAMPLE: Let R be the relation on $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}^+$ consisting of triples (a, b, m) , where a , b , and m are integers with $m \geq 1$ and $a \equiv b \pmod{m}$.

- ❑ $a \bmod m = b \bmod m$, or $(a-b) \bmod m = 0$
- ❑ Then $(8, 2, 3)$, $(-1, 9, 5)$, and $(14, 0, 7)$ all belong to R ,
- ❑ but $(7, 2, 3)$, $(-2, -8, 5)$, and $(11, 0, 6)$ do not belong to R .
- ❑ This relation has **degree** 3.
- ❑ Its first two **domains** are the set of all integers, and its third **domain** is the set of positive integers.

9.2.2 n-ary Relations

EXAMPLE: Let R be the relation consisting of **5-tuples** (A, N, S, D, T) representing **airplane flights**, where A is the airline, N is the flight number, S is the starting point, D is the destination, and T is the departure time.

- ❑ if Nadir Express Airlines has flight 963 from Newark to Bangor at 15:00, then $(\text{Nadir}, 963, \text{Newark}, \text{Bangor}, 15:00)$ belongs to R .
- ❑ The **degree** of this relation is 5.
- ❑ Its **domains** are the set of all airlines, the set of flight numbers, the set of cities, the set of cities (again), and the set of times.

9.2.3 Databases and Relations

The **time required** to manipulate information in a database depends on how this information is stored.

The operations of **adding** and **deleting** records, **updating** records, searching for records, and combining records from overlapping databases are performed millions of times each day in a large database.

Because of the importance of these operations, **various methods** for representing databases have been developed. We will discuss one of these methods, called the **relational data model**, based on the concept of a relation.

MySQL, Microsoft SQL Server, Microsoft Access, Oracle

Records and fields

A database consists of **records**, which are n-tuples, made up of **fields**. The fields are the entries of the n-tuples.

For instance, a database of **student records** may be made up of **fields** containing the name, student number, major, and grade point average of the student.

The relational data model represents a **database** of records as an **n-ary relation**.

A sample database of six such records is:

(Ackermann, 231455, Computer Science, 3.88)

(Adams, 888323, Physics, 3.45)

(Chou, 102147, Computer Science, 3.49)

(Goodfriend, 453876, Mathematics, 3.45)

(Rao, 678543, Mathematics, 3.90)

(Stevens, 786576, Psychology, 2.99).

Tables

Relations used to represent databases are also called **tables**.

Each **column** of the table corresponds to an attribute of the database.

TABLE 1 Students.			
<i>Student_name</i>	<i>ID_number</i>	<i>Major</i>	<i>GPA</i>
Ackermann	231455	Computer Science	3.88
Adams	888323	Physics	3.45
Chou	102147	Computer Science	3.49
Goodfriend	453876	Mathematics	3.45
Rao	678543	Mathematics	3.90
Stevens	786576	Psychology	2.99

Primary key

A domain of an n -ary relation is called a **primary key** when the value of the n -tuple from this domain **determines** the n -tuple.

That is, a domain is a primary key when no two n -tuples in the relation have the same value from this domain.

Records are often added to or deleted from databases. Because of this, the property that a domain is a primary key is time-dependent.

Consequently, a primary key **should be chosen** that remains one whenever the database is changed.

Primary key

 <i>Student_name</i>	 <i>ID_number</i>	 <i>Major</i>	 <i>GPA</i>
Ackermann	231455	Computer Science	3.88
Adams	888323	Physics	3.45
Chou	102147	Computer Science	3.49
Goodfriend	453876	Mathematics	3.45
Rao	678543	Mathematics	3.90
Stevens	786576	Psychology	2.99

Composite key

Combinations of domains can also uniquely identify n-tuples in an n-ary relation.

When the values of a set of domains determine an n-tuple in a relation, the Cartesian product of these domains is called a **composite key**.

<i>Student_name</i>	<i>ID_number</i>	<i>Major</i>	<i>GPA</i>
Ackermann	231455	Computer Science	3.88
Adams	888323	Physics	3.45
Chou	102147	Computer Science	3.49
Goodfriend	453876	Mathematics	3.45
Rao	678543	Mathematics	3.90
Stevens	786576	Psychology	2.99

9.2.4 Operations on n-ary Relations

There are a variety of **operations on n-ary relations** that can be used to form new n-ary relations.

The most basic operation on an n-ary relation is determining all n-tuples in the n-ary relation that **satisfy certain conditions**.

- ❑ We may want to find all the records of all computer science majors in a database of student records.
- ❑ We may want to find all students who have a grade point average above 3.5.
- ❑ We may want to find the records of all computer science majors who have a grade point average above 3.5.

9.2.4 Operations on n-ary Relations

DEFINITION 2: Let R be an n -ary relation and C a condition that elements in R may satisfy. Then the selection operator σ_C maps the n -ary relation R to the n -ary relation of all n -tuples from R that satisfy the condition C .

<i>Student_name</i>	<i>ID_number</i>	<i>Major</i>	<i>GPA</i>
Ackermann	231455	Computer Science	3.88
Adams	888323	Physics	3.45
Chou	102147	Computer Science	3.49
Goodfriend	453876	Mathematics	3.45
Rao	678543	Mathematics	3.90
Stevens	786576	Psychology	2.99

Operations on n-ary Relations

<i>Student_name</i>	<i>ID_number</i>	<i>Major</i>	<i>GPA</i>
Ackermann	231455	Computer Science	3.88
Adams	888323	Physics	3.45
Chou	102147	Computer Science	3.49
Goodfriend	453876	Mathematics	3.45
Rao	678543	Mathematics	3.90
Stevens	786576	Psychology	2.99

The operator s_{C_1} , where C_1 is the condition **Major="Computer Science."**

□ (Ackermann, 231455, Computer Science, 3.88)

□ (Chou, 102147, Computer Science, 3.49)

We use the operator s_{C_2} , where C_2 is the condition **GPA > 3.5.**

□ (Ackermann, 231455, Computer Science, 3.88)

□ (Rao, 678543, Mathematics,

Projections

Projections are used to form new n-ary relations by deleting the same fields in every record of the relation.

DEFINITION 3: The **projection** $P_{i_1 i_2, \dots, i_m}$ where $i_1 < i_2 < \dots < i_m$, maps the n-tuple $(a_1, a_2, \dots, a_n)$ to the m-tuple $(a_{i_1}, a_{i_2}, \dots, a_{i_m})$, where $m \leq n$.

In other words, the projection $P_{i_1 i_2, \dots, i_m}$ **deletes** $n - m$ of the components of an n-tuple, leaving the i_1 th, i_2 th, ..., and i_m th components.

EXAMPLE: What results when the projection $P_{1,3}$ is applied to the:

$(2, 3, 0, 4)$ $(2, 0)$

$(\text{Jane Doe}, 234111001, \text{Geography}, 3.14)$ $(\text{Jane Doe}, \text{Geography})$

$(a_1, a_2, a_3, a_4)?$ (a_1, a_3)

Projections

How **new relations** are produced using projections?

<i>Student_name</i>	<i>ID_number</i>	<i>Major</i>	<i>GPA</i>
Ackermann	231455	Computer Science	3.88
Adams	888323	Physics	3.45
Chou	102147	Computer Science	3.49
Goodfriend	453876	Mathematics	3.45
Rao	678543	Mathematics	3.90
Stevens	786576	Psychology	2.99

$P_{1,4}$



<i>Student_name</i>	<i>GPA</i>
Ackermann	3.88
Adams	3.45
Chou	3.49
Goodfriend	3.45
Rao	3.90
Stevens	2.99

<i>Student</i>	<i>Major</i>	<i>Course</i>
Glauser	Biology	BI 290
Glauser	Biology	MS 475
Glauser	Biology	PY 410
Marcus	Mathematics	MS 511
Marcus	Mathematics	MS 603
Marcus	Mathematics	CS 322
Miller	Computer Science	MS 575
Miller	Computer Science	CS 455

$P_{1,2}$



<i>Student</i>	<i>Major</i>
Glauser	Biology
Marcus	Mathematics
Miller	Computer Science

fewer rows after this projection.

Join operation

The **join operation** is used to combine two tables into one when these tables share some identical fields.

DEFINITION 4: Let R be a relation of degree m and S a relation of degree n . The join $J_p(R, S)$, where $p \leq m$ and $p \leq n$, is a relation of degree $m + n - p$ that consists of all $(m + n - p)$ -tuples $(a_1, a_2, \dots, a_{m-p}, c_1, c_2, \dots, c_p, b_1, b_2, \dots, b_{n-p})$, where the m -tuple $(a_1, a_2, \dots, a_{m-p}, c_1, c_2, \dots, c_p)$ belongs to R and the n -tuple $(c_1, c_2, \dots, c_p, b_1, b_2, \dots, b_{n-p})$ belongs to S .

Join operation

<i>Professor</i>	<i>Department</i>	<i>Course_</i> <i>number</i>
Cruz	Zoology	335
Cruz	Zoology	412
Farber	Psychology	501
Farber	Psychology	617
Grammer	Physics	544
Grammer	Physics	551
Rosen	Computer Science	518
Rosen	Mathematics	575



<i>Department</i>	<i>Course_</i> <i>number</i>	<i>Room</i>	<i>Time</i>
Computer Science	518	N521	2:00 P.M.
Mathematics	575	N502	3:00 P.M.
Mathematics	611	N521	4:00 P.M.
Physics	544	B505	4:00 P.M.
Psychology	501	A100	3:00 P.M.
Psychology	617	A110	11:00 A.M.
Zoology	335	A100	9:00 A.M.
Zoology	412	A100	8:00 A.M.



J_2

<i>Professor</i>	<i>Department</i>	<i>Course_number</i>	<i>Room</i>	<i>Time</i>
Cruz	Zoology	335	A100	9:00 A.M.
Cruz	Zoology	412	A100	8:00 A.M.
Farber	Psychology	501	A100	3:00 P.M.
Farber	Psychology	617	A110	11:00 A.M.
Grammer	Physics	544	B505	4:00 P.M.
Rosen	Computer Science	518	N521	2:00 P.M.
Rosen	Mathematics	575	N502	3:00 P.M.

9.2.5 SQL

The database query language **SQL** (short for **Structured Query Language**) can be used to carry out the operations we have described in this section.

select Departure_time **from** Flights **where** Destination = 'Detroit'

<i>Airline</i>	<i>Flight_number</i>	<i>Gate</i>	<i>Destination</i>	<i>Departure_time</i>
Nadir	122	34	Detroit	08:10 ✓
Acme	221	22	Denver	08:17
Acme	122	33	Anchorage	08:22
Acme	323	34	Honolulu	08:30
Nadir	199	13	Detroit	08:47 ✓
Acme	222	22	Denver	09:10
Nadir	322	34	Detroit	09:44 ✓

SQL

`select Professor, Time from Teaching_assignments, Class_schedule where Teaching_assignments.Department = Class_schedule .Department and Department='Mathematics'`

TABLE 5 Teaching_assignments.

<i>Professor</i>	<i>Department</i>	<i>Course_ number</i>
Cruz	Zoology	335
Cruz	Zoology	412
Farber	Psychology	501
Farber	Psychology	617
Grammer	Physics	544
Grammer	Physics	551
Rosen	Computer Science	518
Rosen	Mathematics	575

TABLE 6 Class_schedule.

<i>Department</i>	<i>Course_ number</i>	<i>Room</i>	<i>Time</i>
Computer Science	518	N521	2:00 P.M.
Mathematics	575	N502	3:00 P.M.
Mathematics	611	N521	4:00 P.M.
Physics	544	B305	4:00 P.M.
Psychology	501	A100	3:00 P.M.
Psychology	617	A110	11:00 A.M.
Zoology	335	A100	9:00 A.M.
Zoology	412	A100	8:00 A.M.

The output would consist of the single 2-tuple (Rosen, 3:00 P.M.).

SQL



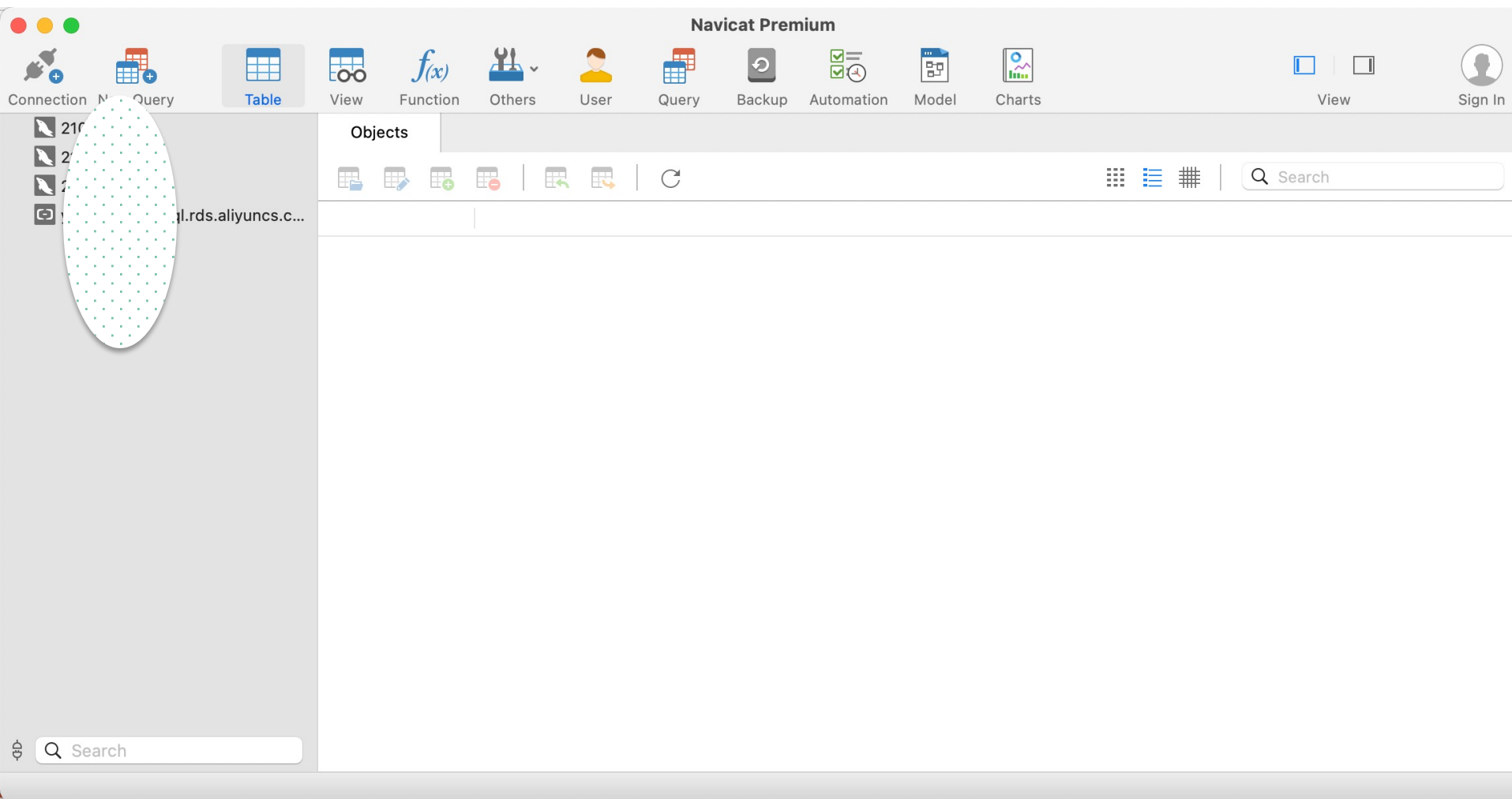
Navicat
Premium

15.0.22 - Premium

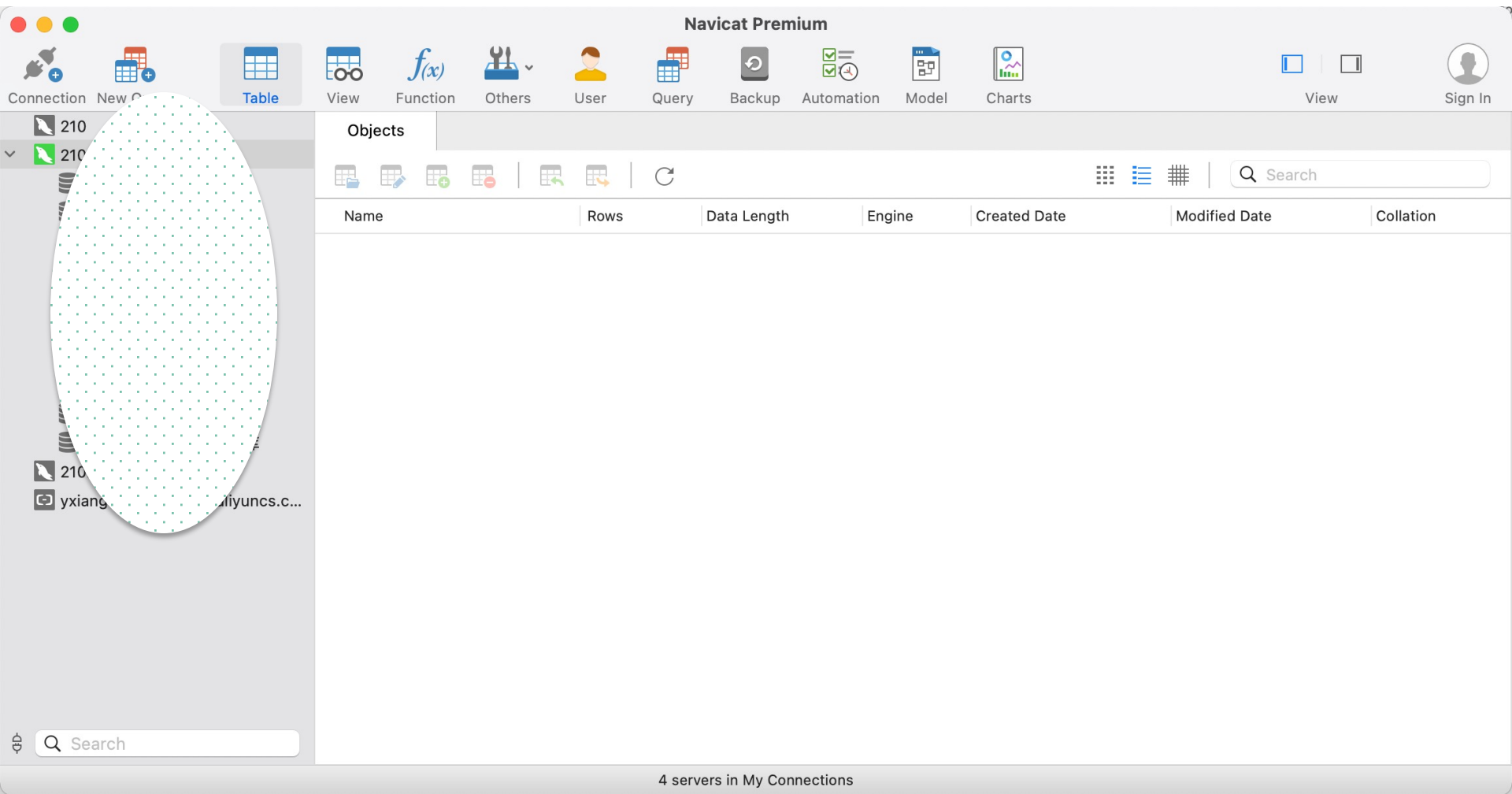


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SQL



SQL



SQL

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Connection New Query Table View Function Others User Query Backup Automation Model Charts View Sign In

210
210.73.28.6

- data_show_lidar
- information_schema
- magge_ae33
- mysql
- opc
- performance_schema
- pollution_data_show**
 - Tables
 - Views
 - Functions
 - Events
 - Queries
 - Backups
- seorae
- test
- 大气参数立体探测数据库

21022222
yxiang818.mysql.rds.aliyuncs.c...

Objects

Name	Rows	Data Length	Engine	Created Date	Modified Date	Collation
data_12km_da	0	16.00 KB	InnoDB	2020-04-08 16:26:41		utf8_general_ci
data_12km_noda	0	16.00 KB	InnoDB	2020-04-08 16:26:41		utf8_general_ci
data_36km_noda	0	16.00 KB	InnoDB	2020-04-08 16:26:41		utf8_general_ci
data_gkd	89,299,446	14446368.00 KB	InnoDB	2020-04-09 15:56:15		utf8_general_ci
data_gkd2	94,297,752	13262624.00 KB	InnoDB	2020-04-08 16:26:41		utf8_general_ci
data_lidar_da	0	16.00 KB	InnoDB	2020-04-08 16:26:42		utf8_general_ci
data_sim	1,495,179	307184.00 KB	InnoDB	2021-08-09 18:13:57		utf8_general_ci
data_sim_ep	0	16.00 KB	InnoDB	2020-04-08 16:26:42		utf8_general_ci
india_data	123,850	24112.00 KB	InnoDB	2020-04-08 16:26:42		utf8_general_ci
india_data_sim	892,828	134144.00 KB	InnoDB	2020-04-08 16:26:42		utf8_general_ci
india_station	157	16.00 KB	InnoDB	2020-04-08 16:26:42		utf8_general_ci
sheet3	1	16.00 KB	InnoDB	2020-04-08 16:26:42		utf8_general_ci
station	2,165	192.00 KB	InnoDB	2021-02-05 17:17:33		utf8_general_ci
station_20210205	1,597	144.00 KB	InnoDB	2021-02-05 17:18:01		utf8_general_ci

14 tables, 14 in current group

SQL

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Connection Manager Query Table View Function $f(x)$ Others User Query Backup Automation Model Charts View Sign In

Objects data_gkd@pollution_dat...

id	datetime	ilevel	height	gkd_aqi	gkd_pm25	gkd_pm10	gkd_o3	gkd_no2	gkd_so2	gkd_co
1001	2014-05-13 00:00:00	0	0	73	41	96	92	42	17	0.7
1001	2014-05-13 01:00:00	0	0	74	40	97	74	51	14	0.7
1001	2014-05-13 02:00:00	0	0	76	39	101	64	51	14	0.6
1001	2014-05-13 03:00:00	0	0	76	38	102	37	67	7	0.7

Objects data_gkd@pollution_dat... station@pollution_data_...

id	stationID	stationName	stationCity	stationLon	stationLat	stationCompare
1001	1001A	万寿西宫	北京	116.3621	39.8784	N
1001	1002	定陵 (对照点)	北京	116.2202	40.2915	Y
1001	1003	东四	北京	116.4174	39.9289	N
1001	1004	天坛	北京	116.4072	39.8863	N
1001	1005	农展馆	北京	116.462	39.9365	N
1001	1006	官园	北京	116.3392	39.9295	N
1001	1007	海淀万柳	北京	116.2878	39.9611	N
1001	1008	顺义新城	北京	116.6636	40.135	N
1001	1009	怀柔镇	北京	116.6275	40.3275	N
1001	1010	昌平镇	北京	116.2296	40.217	N
1001	1011	奥体中心	北京	116.3966	39.9821	N
1001	1012	古城	北京	116.184	39.9136	N
1001	1013	市监测中心 (停运150702)	天津	117.151	39.097	Y
1001	1014	南口路 (停运161122)	天津	117.193	39.173	Y

SELECT * FROM `pollution\_data\_show`.`data\_gkd` LIMIT...

1000 records in page 1

SQL

Fields								Indexes	Foreign Keys	Triggers	Options	Comment	SQL Preview
Name	Type	Length	Decimals	Not Null	Key	Comment							
id	int	11	0	☒	 1								
datetime	datetime	0	0	☒	 2								
ilevel	int	3	0	☒									
height	double	8	0	☒									
gkd_aqi	varchar	255	0	☐									
gkd_pm25	varchar	255	0	☐									
gkd_pm10	varchar	255	0	☐									
gkd_o3	varchar	255	0	☐									
gkd_no2	varchar	255	0	☐									
gkd_so2	varchar	255	0	☐									
gkd_co	varchar	255	0	☐									

	Fields	Indexes	Foreign Keys	Triggers	Options	Comment	SQL Preview
Name	Type	Length	Decimals	Not Null	Key	Comment	
id	int	11	0	☒	 1		
stationID	varchar	255	0	☒			
stationName	varchar	255	0	☒			
stationCity	varchar	255	0	☒			
stationLon	varchar	255	0	☐			
stationLat	varchar	255	0	☐			
stationCompare	varchar	5	0	☐			

SQL

210.73.28.6

pollution_data_...

▶

■

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```
1 SELECT
2   station.stationName,station.stationLat,station.stationLon,data_gkd.datetime,data_gkd.gkd_aqi,
3   data_gkd.gkd_pm25,data_gkd.gkd_pm10,data_gkd.gkd_o3,data_gkd.gkd_no2,data_gkd.gkd_so2,data_gkd.gkd_co
4 FROM
5   data_gkd,station
6 WHERE
7   station.id = data_gkd.id AND station.stationCity = '杭州'
8   AND data_gkd.datetime >= '2016-08-20 00:00:00' AND data_gkd.datetime <= '2016-09-10 00:00:00'
```

MessageResult 1ProfileStatus

stationName	stationLat	stationLon	datetime	gkd_aqi	gkd_pm25	gkd_pm10	gkd_o3	gkd_no2	gkd_so2	gkd_co
四溪	30.2747	120.0633	2016-08-20 00:00:00	37	24	37	55	25	7	0.9
千岛湖(对照点)	29.635	119.026	2016-08-20 00:00:00	26	18	21	48	5	3	0.2
下沙	30.3153	120.3461	2016-08-20 00:00:00	57	38	63	38	67	4	0.5
卧龙桥	30.2456	120.1269	2016-08-20 00:00:00	38	(NULL)	38	31	24	6	0.7
浙江农大	30.2692	120.1903	2016-08-20 00:00:00	41	28	41	47	28	4	0.6
朝晖五区	30.2897	120.157	2016-08-20 00:00:00	43	30	43	46	42	9	0.8
和睦小学	30.3119	120.1197	2016-08-20 00:00:00	43	30	40	46	17	9	0.5
临平镇	30.4183	120.3019	2016-08-20 00:00:00	58	41	57	73	34	9	0.8
城厢镇	30.1819	120.2697	2016-08-20 00:00:00	54	38	55	55	44	3	0.6
云栖	30.1808	120.0883	2016-08-20 00:00:00	46	32	45	13	43	8	0.4
滨江	30.2111	120.2072	2016-08-20 01:00:00	44	25	44	58	29	8	0.7
西溪	30.2747	120.0633	2016-08-20 01:00:00	36	24	36	45	29	7	0.9
千岛湖(对照点)	29.635	119.026	2016-08-20 01:00:00	28	19	22	43	5	4	0.2

+ - ✓ ✕

SELECT station.stationName, station.stationLat, station.stationLon, data_gkd.datetime, data_gkd.gk...

37.627s elapsed

🔍 📄

SQL

210.73.28.6

pollution_data_...

1

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5

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8

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SELECT

station.stationName,station.stationLat,station.stationLon,

data_gkd.datetime,data_gkd.gkd_aqi,data_gkd.gkd_pm25,data_gkd.gkd_pm10,data_gkd.gkd_o3,

data_gkd.gkd_no2,data_gkd.gkd_so2,data_gkd.gkd_co

FROM

data_gkd,station

WHERE

station.id = data_gkd.id AND data_gkd.datetime = '2021-08-30 14:00:00'

AND data_gkd.gkd_o3 >= 200 AND data_gkd.gkd_o3 < 2000

Message

Result 1

Profile

Status

stationName	stationLat	stationLon	datetime	gkd_aqi	gkd_pm25	gkd_pm10	gkd_o3	gkd_no2	gkd_so2	gkd_co
草场门	32.0551	118.7538	2021-08-30 14:00:00	107	39	66	214	14	15	1.4
奥体中心	31.9996	118.7356	2021-08-30 14:00:00	111	39	66	221	17	5	1.3
上方山(对照点)	31.2472	120.561	2021-08-30 14:00:00	106	25	61	212	11	6	0.2
安家	31.9087	119.9255	2021-08-30 14:00:00	108	27	41	216	16	4	0.7
坑口子站	23.1522	112.5544	2021-08-30 14:00:00	101	21	35	202	12	9	0.4
吴江市环保局	31.155	120.636	2021-08-30 14:00:00	100	26	41	200	12	9	0.3
海虞子站	31.67	120.758	2021-08-30 14:00:00	103	22	38	206	16	11	0.5
兴福子站	31.671	120.721	2021-08-30 14:00:00	108	22	43	216	15	9	0.5
菱塘子站	31.656	120.734	2021-08-30 14:00:00	106	22	41	212	11	10	0.8

+ - ✓ ✕

SELECT station.stationName, station.stationLat, station.stationLon, data_gkd.datetime, data_gkd.g...

23.945s elapsed

SQL

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Connection New Query Table View Function Event User Query Backup Automation Model Charts View Sign In

210.73 data_gkd@pollution_dat... query_@pollution_data... query_station@pollution... xgtp@seorae (210.73.28... ae33@magge_ae33 (210...

	id	sj	xgxs	tp	station	xgxs532
	19115973	2019-05-15 15:25:03	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.193,0.04	甘肃.白银	0.000,0.000,-0.049,-0.014,-0.051,-0.030,-0.013,0.011,0.040,0.071,0.110,0.153,0.199,0.247,0.297,0.349,0.398,
	19115974	2019-05-15 15:20:...	0.000,0.000,-0.30	-9999.000,-9999.000,0.186,0.191,0.05	甘肃.白银	0.000,0.000,-0.049,-0.014,-0.048,-0.023,-0.001,0.031,0.071,0.115,0.170,0.229,0.291,0.354,0.417,0.480,0.540,
	19115975	2019-05-15 15:15:16	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.191,0.06	甘肃.白银	0.000,0.000,-0.048,-0.009,-0.045,-0.017,0.009,0.046,0.093,0.145,0.208,0.276,0.346,0.417,0.488,0.558,0.62
	19115976	2019-05-15 15:10:23	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.190,0.071	甘肃.白银	0.000,0.000,-0.048,-0.011,-0.044,-0.016,0.011,0.048,0.096,0.150,0.214,0.284,0.357,0.431,0.505,0.577,0.644,
	19115977	2019-05-15 15:05:29	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.190,0.071	甘肃.白银	0.000,0.000,-0.049,-0.010,-0.043,-0.015,0.013,0.052,0.101,0.156,0.222,0.292,0.364,0.435,0.504,0.572,0.635
	19115978	2019-05-15 15:00:35	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.190,0.07	甘肃.白银	0.000,0.000,-0.048,-0.007,-0.041,-0.011,0.018,0.060,0.112,0.170,0.241,0.316,0.395,0.475,0.555,0.633,0.706,
	19115979	2019-05-15 14:55:40	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.189,0.07	甘肃.白银	0.000,0.000,-0.048,-0.011,-0.042,-0.013,0.016,0.057,0.108,0.166,0.235,0.309,0.386,0.464,0.541,0.615,0.684,
	19115980	2019-05-15 14:50:46	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.190,0.071	甘肃.白银	0.000,0.000,-0.049,-0.010,-0.043,-0.014,0.014,0.054,0.104,0.161,0.230,0.305,0.383,0.463,0.542,0.620,0.692,
	19115981	2019-05-15 14:45:53	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.189,0.07	甘肃.白银	0.000,0.000,-0.048,-0.006,-0.041,-0.009,0.022,0.066,0.121,0.183,0.258,0.338,0.422,0.506,0.589,0.670,0.74
	19115982	2019-05-15 14:41:00	0.000,0.000,-0.30	-9999.000,-9999.000,0.186,0.189,0.07	甘肃.白银	0.000,0.000,-0.049,-0.011,-0.042,-0.014,0.015,0.055,0.106,0.163,0.231,0.304,0.378,0.453,0.528,0.601,0.667,
	19115983	2019-05-15 14:36:07	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.189,0.07	甘肃.白银	0.000,0.000,-0.048,-0.009,-0.041,-0.010,0.021,0.064,0.119,0.180,0.253,0.332,0.414,0.497,0.580,0.661,0.736,
	19115984	2019-05-15 14:31:14	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.189,0.07	甘肃.白银	0.000,0.000,-0.048,-0.010,-0.042,-0.012,0.018,0.060,0.113,0.173,0.245,0.322,0.402,0.482,0.562,0.639,0.711,
	19115985	2019-05-15 14:26:21	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.189,0.07	甘肃.白银	0.000,0.000,-0.048,-0.007,-0.040,-0.010,0.021,0.065,0.120,0.182,0.257,0.337,0.421,0.505,0.589,0.669,0.742,
	19115986	2019-05-15 14:21:27	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.190,0.07	甘肃.白银	0.000,0.000,-0.049,-0.009,-0.041,-0.012,0.017,0.059,0.112,0.171,0.242,0.318,0.398,0.478,0.558,0.636,0.707,
	19115987	2019-05-15 14:16:33	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.190,0.07	甘肃.白银	0.000,0.000,-0.049,-0.011,-0.042,-0.012,0.017,0.059,0.112,0.172,0.243,0.320,0.400,0.481,0.561,0.639,0.710,
	19115988	2019-05-15 14:11:39	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.190,0.07	甘肃.白银	0.000,0.000,-0.048,-0.009,-0.041,-0.011,0.019,0.062,0.116,0.177,0.251,0.331,0.414,0.497,0.581,0.661,0.735,
	19115989	2019-05-15 14:06:46	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.189,0.08	甘肃.白银	0.000,0.000,-0.048,-0.010,-0.041,-0.010,0.021,0.064,0.119,0.181,0.255,0.334,0.415,0.496,0.575,0.650,0.718,
	19115990	2019-05-15 14:01:53	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.190,0.07	甘肃.白银	0.000,0.000,-0.048,-0.009,-0.042,-0.011,0.020,0.063,0.119,0.181,0.256,0.336,0.420,0.503,0.586,0.667,0.740,
	19115991	2019-05-15 13:57:00	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.190,0.071	甘肃.白银	0.000,0.000,-0.048,-0.009,-0.043,-0.012,0.019,0.062,0.117,0.179,0.254,0.336,0.420,0.505,0.590,0.672,0.747,
	19115992	2019-05-15 13:52:07	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.190,0.07	甘肃.白银	0.000,0.000,-0.049,-0.011,-0.043,-0.014,0.015,0.056,0.108,0.166,0.237,0.312,0.391,0.470,0.548,0.623,0.692,
	19115993	2019-05-15 13:47:13	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.190,0.06	甘肃.白银	0.000,0.000,-0.049,-0.014,-0.046,-0.019,0.007,0.045,0.092,0.145,0.211,0.281,0.354,0.429,0.503,0.575,0.641,
	19115994	2019-05-15 13:42:19	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.192,0.06	甘肃.白银	0.000,0.000,-0.050,-0.014,-0.047,-0.019,0.007,0.044,0.091,0.144,0.210,0.280,0.354,0.429,0.503,0.575,0.641,
	19115995	2019-05-15 13:37:26	0.000,0.000,-0.30	-9999.000,-9999.000,0.186,0.191,0.06	甘肃.白银	0.000,0.000,-0.050,-0.015,-0.048,-0.022,0.001,0.035,0.077,0.125,0.185,0.249,0.316,0.384,0.453,0.519,0.581,
	19115996	2019-05-15 13:32:33	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.192,0.05	甘肃.白银	0.000,0.000,-0.050,-0.015,-0.048,-0.022,0.001,0.035,0.079,0.129,0.190,0.256,0.324,0.393,0.462,0.528,0.58
	19115997	2019-05-15 13:27:38	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.193,0.04	甘肃.白银	0.000,0.000,-0.050,-0.016,-0.051,-0.027,-0.006,0.024,0.062,0.106,0.160,0.219,0.281,0.345,0.410,0.473,0.533,
	19115998	2019-05-15 13:22:45	0.000,0.000,-0.30	-9999.000,-9999.000,0.186,0.192,0.04	甘肃.白银	0.000,0.000,-0.050,-0.019,-0.052,-0.028,-0.008,0.021,0.059,0.102,0.156,0.214,0.275,0.338,0.400,0.461,0.51
	19115999	2019-05-15 13:17:52	0.000,0.000,-0.30	-9999.000,-9999.000,0.186,0.192,0.04	甘肃.白银	0.000,0.000,-0.049,-0.017,-0.051,-0.027,-0.006,0.024,0.063,0.107,0.162,0.222,0.285,0.349,0.414,0.477,0.53
	19116000	2019-05-15 13:12:59	0.000,0.000,-0.30	-9999.000,-9999.000,0.187,0.192,0.04	甘肃.白银	0.000,0.000,-0.050,-0.019,-0.051,-0.028,-0.008,0.021,0.058,0.101,0.154,0.211,0.272,0.333,0.395,0.456,0.511,
	19116001	2019-05-15 13:08:06	0.000,0.000,-0.30	-9999.000,-9999.000,0.188,0.191,0.03	甘肃.白银	0.000,0.000,-0.049,-0.018,-0.051,-0.027,-0.006,0.024,0.063,0.107,0.163,0.223,0.285,0.350,0.414,0.476,0.53

SELECT * FROM `seorae`.`xgtp` LIMIT 0,1000

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SQL

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Connection New Query Table View Function Event User Query Backup Automation Model Charts View Sign In

Objects data_gkd@pollution_dat... query_@pollution_data... query_station@pollution... xgtp@seorae (210.73.28... ae33@magge_ae33 (210...

Date	Time	Timebase	RefCh1	Sen1Ch1	Sen2Ch1	RefCh2	Sen1Ch2	Sen2Ch2	RefCh3	Sen1Ch3	Sen2Ch3	RefCh4	Sen1Ch4	Sen2Ch4	RefCh5	Sen1Ch
2018-01-29	00:00:00	60	911384	524522	677876	938893	700472	793654	908310	749310	858130	872554	798143	877345	866385	813586
2018-01-29	00:01:00	60	911304	523491	677283	938852	699783	793321	908201	748742	857832	872352	797578	876976	866283	813180
2018-01-29	00:02:00	60	911419	522530	676826	938913	699148	793096	908274	748297	857698	872454	797265	876944	866401	812945
2018-01-29	00:03:00	60	911477	521591	676327	938964	698530	792848	908292	747848	857511	872427	796867	876755	866414	812658
2018-01-29	00:04:00	60	911482	520622	675801	938919	697845	792531	908262	747365	857286	872366	796432	876543	866335	812272
2018-01-29	00:05:00	60	911614	519728	675376	939051	697304	792363	908387	746996	857212	872551	796218	876569	866450	812077
2018-01-29	00:06:00	60	911677	518816	674883	939038	696653	792064	908419	746558	857042	872535	795828	876400	866466	811791
2018-01-29	00:07:00	60	911655	517845	674306	938970	695977	791687	908385	746087	856785	872417	795373	876123	866437	811478
2018-01-29	00:08:00	60	911783	516993	673868	939051	695410	791489	908448	745674	856636	872499	795086	876057	866485	811215
2018-01-29	00:09:00	60	911799	516036	673355	939161	694840	791300	908526	745268	856513	872490	794698	875901	866582	811005
2018-01-29	00:10:00	60	911683	515041	672751	939010	694104	790900	908408	744727	856214	872345	794202	875601	866428	810564
2018-01-29	00:11:00	60	911776	514164	672302	939055	693512	790649	908461	744315	856066	872428	793925	875542	866527	810352
2018-01-29	00:12:00	60	911827	513239	671800	939126	692900	790416	908531	743892	855916	872438	793534	875379	866567	810075
2018-01-29	09:34:00	60	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2018-01-29	09:35:00	60	6200	95	224	6200	25	240	6200	54	224	9200	59	180	6200	54
2018-01-29	09:36:00	60	16900	100	228	6200	35	310	10900	60	232	19700	60	180	9550	60
2018-01-29	09:37:00	60	901731	638905	745258	939119	789070	847897	900248	814752	898975	849890	841139	898637	832384	842881
2018-01-29	09:38:00	60	903343	639838	746476	939783	789563	848404	901094	815467	899757	849922	841143	898357	833217	843604
2018-01-29	09:39:00	60	904337	640390	747203	940198	789847	848710	901562	815836	900159	849891	841052	898129	833658	843938
2018-01-29	09:40:00	60	904860	639243	746935	940429	788887	848387	901770	815113	899966	849843	840230	897657	833869	843470
2018-01-29	09:41:00	60	905252	635728	745686	940657	786277	847517	901998	813115	899314	849747	838282	896814	834021	842031
2018-01-29	09:42:00	60	905609	632212	744381	940806	783591	846552	902100	810996	898545	849596	836272	895898	834108	840536
2018-01-29	09:43:00	60	905825	628606	742958	940953	780901	845603	902276	808923	897829	849562	834364	895144	834236	839065
2018-01-29	09:44:00	60	906020	625012	741510	941006	778148	844554	902364	806804	897059	849422	832376	894281	834334	837586
2018-01-29	09:45:00	60	906245	621468	740080	941120	775434	843582	902481	804687	896293	849374	830453	893504	834371	836024
2018-01-29	09:46:00	60	906394	617909	738597	941195	772697	842553	902526	802513	895458	849180	828396	892597	834420	834492
2018-01-29	09:47:00	60	906536	614383	737112	941203	769954	841494	902611	800381	894667	849046	826412	891747	834435	832913
2018-01-29	09:48:00	60	906695	610897	735648	941309	767243	840479	902658	798232	893835	848889	824378	890861	834468	831362
2018-01-29	09:49:00	60	906874	607441	734193	941380	764520	839445	902729	796076	893042	848811	822437	890074	834502	829818

SELECT * FROM `magge\_ae33`.`ae33` LIMIT 0,1000

1000 records in page 1

SQL

Navicat Premium

Connection New Query Table View Function Event User Query Backup Automation Model Charts View Sign In

Objects << xgtp@seoriae (210.73.28... ae33@magge_ae33 (210... fuzzytable_fmksdb@opc... fuzzytable_ytwnl@opc (... data@opc (210.73.28.6)

dt	id	wll	ywwml	ytwnl	fjckwd	yswd	hcydl	ycfpl	co	nox	fjlzwd	fuzzy_ytwnl	fuzzy_fmksdb
2021-05-29 00:00:03	一期	222.8938	12.95263	6.129671	887.033	1033.553	643.5165	1402.657	0.04053724	37.36264	841.6118	(NULL)	(NULL)
2021-05-29 00:00:09	一期	222.8938	12.95263	6.116484	886.7399	1033.846	574.3589	1401.573	0.03907204	37.39927	841.6118	(NULL)	(NULL)
2021-05-29 00:00:16	一期	222.8938	12.94908	6.136264	886.1539	1033.553	654.652	1401.934	0.03907204	37.39927	841.9048	(NULL)	(NULL)
2021-05-29 00:00:24	一期	222.8327	12.95263	6.127473	885.8608	1033.26	620.6594	1401.573	0.04004884	37.47253	841.3187	(NULL)	(NULL)
2021-05-29 00:00:31	一期	222.9548	12.95263	6.140659	885.2747	1033.26	659.9268	1402.296	0.04053724	37.54579	841.0256	(NULL)	(NULL)
2021-05-29 00:00:37	一期	222.9548	12.94908	6.129671	885.2747	1033.26	592.5275	1402.657	0.04004884	37.58242	841.6118	(NULL)	(NULL)
2021-05-29 00:00:44	一期	222.9548	12.95263	6.123077	884.3956	1033.26	581.978	1402.296	0.04004884	37.58242	841.3187	(NULL)	(NULL)
2021-05-29 00:00:51	一期	222.8938	12.95263	6.127473	883.8096	1033.26	608.3517	1401.934	0.04004884	37.61905	841.3187	(NULL)	(NULL)
2021-05-29 00:00:57	一期	222.8938	12.95263	6.129671	882.9304	1032.967	590.7692	1402.657	0.04004884	37.65568	841.3187	(NULL)	(NULL)
2021-05-29 00:01:04	一期	222.8938	12.95263	6.134066	882.6374	1032.674	627.6923	1401.934	0.03956044	37.65568	840.7326	(NULL)	(NULL)
2021-05-29 00:01:11	一期	222.9548	12.95263	6.129671	882.3444	1032.381	621.8315	1402.657	0.04004884	37.65568	841.0256	(NULL)	(NULL)
2021-05-29 00:01:18	一期	222.8327	12.95263	6.123077	881.7582	1032.674	632.381	1401.573	0.03956044	37.65568	840.7326	(NULL)	(NULL)
2021-05-29 00:01:25	一期	222.8327	12.95263	6.125275	881.7582	1032.674	553.8461	1402.657	0.03907204	37.61905	840.4396	(NULL)	(NULL)
2021-05-29 00:01:32	一期	222.8938	12.94908	6.134066	881.4652	1032.674	672.8206	1401.573	0.03956044	37.61905	840.1465	(NULL)	(NULL)
2021-05-29 00:01:39	一期	222.8938	12.95263	6.127473	880.5861	1032.674	564.9817	1402.296	0.04004884	37.61905	839.8535	(NULL)	(NULL)
2021-05-29 00:01:46	一期	222.8938	12.94908	6.140659	880.8792	1032.088	670.4762	1401.573	0.04053724	37.58242	839.5605	(NULL)	(NULL)
2021-05-29 00:01:52	一期	222.8327	12.95263	6.134066	880.5861	1032.088	492.8938	1402.296	0.04249084	37.50916	839.5605	(NULL)	(NULL)
2021-05-29 00:01:59	一期	222.8327	12.94908	6.138462	880.293	1031.795	656.9964	1402.296	0.04444445	37.50916	841.6118	(NULL)	(NULL)
2021-05-29 00:02:06	一期	222.8938	12.95263	6.134066	879.707	1031.795	521.6118	1401.934	0.04542125	37.47253	841.3187	(NULL)	(NULL)
2021-05-29 00:02:12	一期	222.8327	12.95263	6.125275	879.4139	1031.795	648.2051	1402.657	0.04590965	37.50916	840.7326	(NULL)	(NULL)
2021-05-29 00:02:19	一期	222.8327	12.94908	6.129671	879.1209	1031.502	512.8205	1402.296	0.04590965	37.50916	840.4396	(NULL)	(NULL)
2021-05-29 00:02:27	一期	222.8938	12.95263	6.134066	878.5348	1031.502	710.9158	1401.573	0.04835165	37.54579	840.1465	(NULL)	(NULL)
2021-05-29 00:02:34	一期	222.8327	12.95263	6.120879	878.5348	1031.209	533.3333	1402.657	0.05128205	37.61905	839.8535	(NULL)	(NULL)
2021-05-29 00:02:40	一期	222.8938	12.95263	6.129671	878.2418	1031.209	691.5751	1402.296	0.05372405	37.65568	839.8535	(NULL)	(NULL)
2021-05-29 00:02:47	一期	222.8327	12.95263	6.120879	878.5348	1031.209	559.1209	1402.657	0.05372405	37.72894	839.2674	(NULL)	(NULL)
2021-05-29 00:02:54	一期	222.8938	12.94908	6.129671	878.2418	1031.209	671.0623	1402.657	0.05372405	37.72894	839.2674	(NULL)	(NULL)
2021-05-29 00:03:00	一期	223.0159	12.95263	6.125275	877.9487	1030.916	552.674	1401.573	0.05470086	37.72894	838.6813	(NULL)	(NULL)
2021-05-29 00:03:07	一期	222.9548	12.95263	6.125275	877.6557	1030.916	683.9561	1402.296	0.05616606	37.65568	838.6813	(NULL)	(NULL)
2021-05-29 00:03:14	一期	222.8938	12.95617	6.127473	877.6557	1030.916	526.3004	1402.657	0.05860806	37.61905	838.3883	(NULL)	(NULL)

SELECT * FROM `opc`.`data` LIMIT 0,1000

1000 records in page 1

9.2.6 Association Rules from Data Mining

Data mining, the discipline with the goal of gaining useful information from data.

Transaction, a set of items bought by a customer during a visit to the store, such as {milk, eggs, bread} or {orange juice, bananas, yogurt, cream}.

We call each product in the store an **item**.

A collection of items is known as an **itemset** (also called transaction, or basket).

A **k-itemset** is an itemset that contains exactly k items.

Association Rules from Data Mining

DEFINITION 5: The **count** of an itemset I , denoted by $\sigma(I)$, in a set of transactions $T = \{t_1, t_2, \dots, t_k\}$, where k is a positive integer, is the number of transactions that contain this itemset. That is,

$$\sigma(I) = |\{t_i \in T \mid I \subseteq t_i\}|.$$

The **support** of an itemset I is the **probability** that I is included in a randomly selected transaction from T . That is,

$$\text{support}(I) = \frac{\sigma(I)}{|T|}.$$

The **support threshold** s is specified for a particular application.

Frequent itemset mining is the process of finding itemset I with support greater than or equal to s . Such itemset are said to be **frequent**.

Association Rules from Data Mining

TABLE 9 A Set of Transactions.

<i>Transaction Number</i>	<i>Items</i>
1	{apples, pears, mangos}
2	{pears, cider}
3	{apples, cider, donuts, mangos}
4	{apples, pears, cider, donuts}
5	{apples, cider, donuts}
6	{pears, cider, donuts}
7	{pears, donuts}
8	{apples, pears, cider}

$$\sigma(\{\text{apples}\}) = 5$$

$$\text{support}(\{\text{apples}\}) = 5/8$$

$$\sigma(\{\text{apples, cider}\}) = 4$$

$$\text{support}(\{\text{cider}\}) = 4/8 = 1/2.$$

TABLE 10 Binary Database for the Transactions in Table 9.

<i>Transaction Number</i>	<i>Apples</i>	<i>Pears</i>	<i>Cider</i>	<i>Donuts</i>	<i>Mangos</i>
1	1	1	0	0	1
2	0	1	1	0	0
3	1	0	1	1	1
4	1	1	1	1	0
5	1	0	1	1	0
6	0	1	1	1	0
7	0	1	0	1	0
8	1	1	1	0	0

Association Rules from Data Mining

TABLE 10 Binary Database for the Transactions in Table 9.

<i>Transaction Number</i>	<i>Apples</i> ✓	<i>Pears</i> ✓	<i>Cider</i> ✓	<i>Donuts</i> ✓	<i>Mangos</i>
1	1	1	0	0	1
2	0	1	1	0	0
3	1	0	1	1	1
4	1	1	1	1	0
5	1	0	1	1	0
6	0	1	1	1	0
7	0	1	0	1	0
8	1	1	1	0	0

If we set the **support threshold** to be 0.5 (4/8), an item is frequent if it occurs in at least four of the eight transactions.

- ❑ apples, pears, cider and donuts are the frequent items.
- ❑ The itemset {apples, cider} is a frequent itemset.
- ❑ But the itemset {donuts, pears} is not a frequent itemset.

Association Rules from Data Mining

If S is a set of items and T is a set of transactions, an association rule is an implication of the form $I \rightarrow J$, where I and J are disjoint subsets of S . Although this notation borrows the notation for logical implications, its meaning is not entirely analogous.

DEFINITION 6: If I and J are subsets of a set T of transactions, then

$$\text{support}(I \rightarrow J) = \frac{\sigma(I \cup J)}{|T|}$$

and

$$\text{confidence}(I \rightarrow J) = \frac{\sigma(I \cup J)}{\sigma(I)}.$$

Association Rules from Data Mining

What are the **support** and the **confidence** of the association rule $\{\text{cider, donuts}\} \rightarrow \{\text{apples}\}$ for the set of transactions?

TABLE 10 Binary Database for the Transactions in Table 9.

Transaction Number	Apples ✓	Pears	Cider ✓	Donuts ✓	Mangos
1	1	1	0	0	1
2	0	1	1	0	0
3	1	0	1	1	1
4	1	1	1	1	0
5	1	0	1	1	0
6	0	1	1	1	0
7	0	1	0	1	0
8	1	1	1	0	0

$$\text{support}(I \rightarrow J) = \frac{\sigma(I \cup J)}{|T|}$$

$$\text{confidence}(I \rightarrow J) = \frac{\sigma(I \cup J)}{\sigma(I)}$$

$\sigma(\{\text{cider, donuts}\} \cup \{\text{apples}\}) = \sigma(\{\text{cider, donuts, apples}\}) = 3$ and $|T| = 8$

$$\text{support}(I \rightarrow J) = 3/8 = 0.375$$

$$\text{confidence}(I \rightarrow J) = \sigma(\{\text{cider, donuts}\} \cup \{\text{apples}\}) / \sigma(\{\text{cider, donuts}\}) = 3/4 = 0.75$$

Section Summary

n-ary Relations

domains, degree

Databases and Relations

Records, Tables, Primary key, Composite key

Operations on n-ary Relations

projection, Join operation

SQL

Association Rules from Data Mining

Homework after class

9.2 n-ary Relations and Their Applications

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Representing Relations

Section 9.3

9.3.1 Introduction

There are many ways to represent a relation between finite sets.

- ❑ List of ordered pairs.
- ❑ A table
- ❑ zero–one matrices
- ❑ directed graphs

Generally, **matrices** are appropriate for the representation of relations in **computer programs**.

On the other hand, people often find the representation of relations using **directed graphs** useful for **understanding the properties** of these relations.

9.3.2 Representing Relations Using Matrices

A **relation** between **finite sets** can be represented using a **zero-one matrix**.

Suppose R is a relation from $A = \{a_1, a_2, \dots, a_m\}$ to $B = \{b_1, b_2, \dots, b_n\}$.

- The elements of the two sets can be listed in any particular arbitrary order. When $A = B$, we use the same ordering.

The relation R is represented by the matrix $M_R = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R, \\ 0 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

The matrix representing R has a 1 as its (i,j) entry when a_i is related to b_j and a 0 if a_i is not related to b_j .

EXAMPLES of Representing Relations

Using Matrices₁

EXAMPLE: Suppose that $A = \{1,2,3\}$ and $B = \{1,2\}$. Let R be the relation from A to B containing (a,b) if $a \in A$, $b \in B$, and $a > b$. What is the matrix representing R (assuming the ordering of elements is the same as the increasing numerical order)?

Solution: Because $R = \{(2,1), (3,1),(3,2)\}$, the matrix is

$$M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

EXAMPLEs of Representing Relations

Using Matrices₂

EXAMPLE: Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4, b_5\}$. Which **ordered pairs** are in the relation R represented by the matrix

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} ?$$

Solution: Because R consists of those ordered pairs (a_i, b_j) with $m_{ij} = 1$, it follows that:

$$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3), (a_3, b_5)\}.$$

Matrices of Relations on Sets

If R is a **reflexive** relation, all the elements on the main diagonal of M_R are equal to 1.

$$\begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & \ddots & & \\ & & & & \ddots & \\ & & & & & 1 \\ & & & & & & 1 \end{bmatrix}$$

R is a **symmetric** relation, if and only if $m_{ij} = 1$ whenever $m_{ji} = 1$.

$$\begin{bmatrix} & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & \ddots & & \\ & & & & \ddots & \\ & & & & & 1 \\ & & & & & & 1 \end{bmatrix}$$

R is an **antisymmetric** relation, if and only if $m_{ij} = 0$ or $m_{ji} = 0$ when $i \neq j$.

$$\begin{bmatrix} & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & \ddots & & \\ & & & & \ddots & \\ & & & & & 1 \\ & & & & & & 1 \end{bmatrix}$$

EXAMPLE of a Relation on a Set

EXAMPLE: Suppose that the relation R on a set is represented by the matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Is R reflexive, symmetric, and/or antisymmetric?

Solution: Because all the diagonal elements are equal to 1, R is **reflexive**.

Because M_R is symmetric, R is **symmetric** and not **antisymmetric** because both $m_{1,2}$ and $m_{2,1}$ are 1.

Union and Intersection of Relations

Suppose that R_1 and R_2 are relations on a set A represented by the matrices M_{R_1} and M_{R_2} , respectively.

The matrix representing the **union** of these relations has a 1 in the positions where either M_{R_1} or M_{R_2} has a 1.

$$M_{R_1 \cup R_2} = M_{R_1} \vee M_{R_2}$$

The matrix representing the **intersection** of these relations has a 1 in the positions where both M_{R_1} and M_{R_2} have a 1.

$$M_{R_1 \cap R_2} = M_{R_1} \wedge M_{R_2}.$$

Union and Intersection of Relations

Suppose that the relations R_1 and R_2 on a set A are represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

What are the matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$?

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Matrix for the Composite of Relations

Suppose that R is a relation from A to B and S is a relation from B to C . Suppose that A , B , and C have m , n , and p elements, respectively.

Let the zero-one matrices for $S \circ R$, R , and S be $M_{S \circ R} = [t_{ij}]$, $M_R = [r_{ij}]$, and $M_S = [s_{ij}]$, respectively (these matrices have sizes $m \times p$, $m \times n$, and $n \times p$, respectively).

The ordered pair (a_i, c_j) belongs to $S \circ R$ if and only if there is an element b_k such that (a_i, b_k) belongs to R and (b_k, c_j) belongs to S .

It follows that $t_{ij} = 1$ if and only if $r_{ik} = s_{kj} = 1$ for some k .

From the DEFINITION of the Boolean product, this means that

$$M_{S \circ R} = M_R \odot M_S.$$

Recall: Boolean Product of Zero-One Matrices ₁

DEFINITION: Let $\mathbf{A} = [a_{ij}]$ be an $m \times k$ zero-one matrix and $\mathbf{B} = [b_{ij}]$ be a $k \times n$ zero-one matrix. The *Boolean product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} \odot \mathbf{B}$, is the $m \times n$ zero-one matrix with (i,j) th entry

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \dots \vee (a_{ik} \wedge b_{kj}).$$

Note that the Boolean product of $\mathbf{A}$ and $\mathbf{B}$ is obtained in an analogous way to the ordinary product of these matrices, but with **addition** replaced with the operation $\vee$ and with **multiplication** replaced with the operation $\wedge$.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Recall: Boolean Product of Zero-One Matrices ₂

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

$$A \odot B = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \vee & 0 & 1 & \vee & 0 & 0 & \vee & 0 \\ 0 & \vee & 0 & 0 & \vee & 1 & 0 & \vee & 1 \\ 1 & \vee & 0 & 1 & \vee & 0 & 0 & \vee & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Matrix for the Composite of Relations

EXAMPLE: Find the matrix representing the relations $S \circ R$, where the matrices representing R and S are

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \mathbf{M}_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Matrix for the Composite of Relations

The matrix representing the **composite** of two relations can be used to find the matrix for M_{R^n} .

In particular, from the DEFINITION of **Boolean powers**:

$$M_{R^n} = M_R^{[n]}.$$

R^n : Powers of a Relation ($R^n = R^{n-1} \circ R$).

$A^{[r]}$: The r th Boolean power of **A**

$$A^{[r]} = \underbrace{A \odot A \odot \cdots \odot A}_{r \text{ times}}.$$

Matrix for the Composite of Relations

EXAMPLE: Find the matrix representing the relation R^2 , where the matrix representing R is

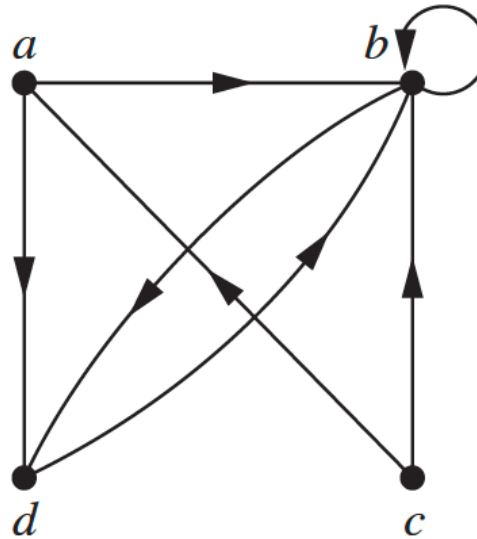
$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

The matrix for R^2 is M_{R^2} .

$$\begin{aligned} M_{R^2} &= M_R^{[2]} \\ &= M_R \odot M_R = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \end{aligned}$$

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \dots \vee (a_{ik} \wedge b_{kj}).$$

9.3.3 Representing Relations Using Digraphs

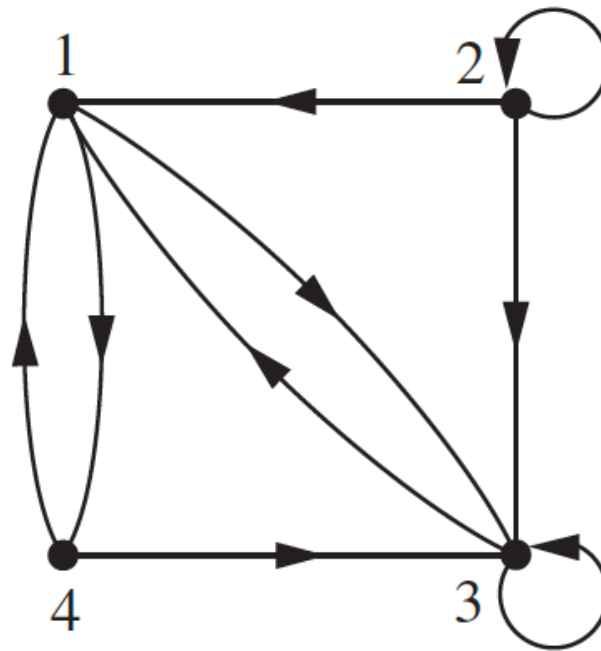


DEFINITION 1: A **directed graph**, or digraph, consists of a set V of **vertices** (or nodes) together with a set E of ordered pairs of elements of V called **edges** (or arcs). The vertex a is called the **initial vertex** of the edge (a,b) , and the vertex b is called the **terminal vertex** of this edge.

- An edge of the form (a, a) is called a **loop**.

EXAMPLES of Digraphs Representing Relations

EXAMPLE: What are the ordered pairs in the relation represented by this directed graph?



Solution: The ordered pairs in the relation are $(1, 3)$, $(1, 4)$, $(2, 1)$, $(2, 2)$, $(2, 3)$, $(3, 1)$, $(3, 3)$, $(4, 1)$, and $(4, 3)$

Determining which Properties a Relation has from its Digraph

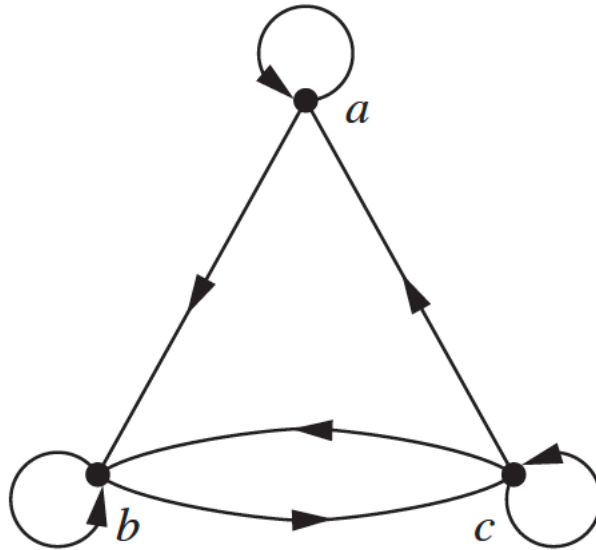
Reflexivity: A loop must be present at all vertices in the graph.

Symmetry: If (x,y) is an edge, then so is (y,x) .

Antisymmetry: If (x,y) with $x \neq y$ is an edge, then (y,x) is not an edge.

Transitivity: If (x,y) and (y,z) are edges, then so is (x,z) .

Determining which Properties a Relation has from its Digraph – EXAMPLE 1



Reflexive?

Yes

Symmetric?

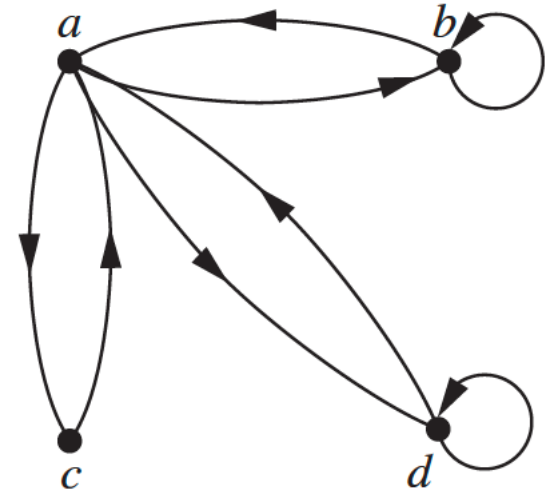
No

Antisymmetric?

No

Transitive?

No



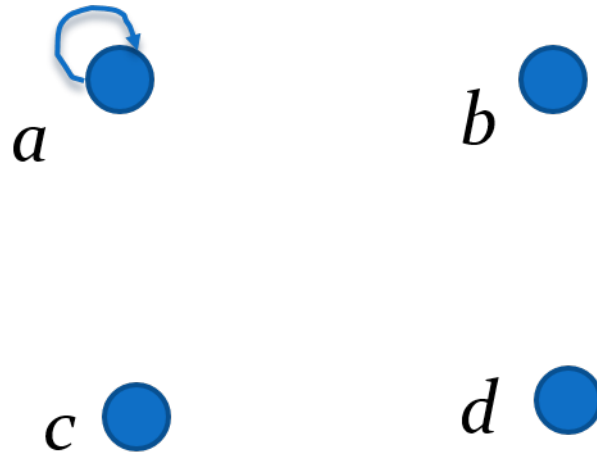
No

Yes

No

No

Determining which Properties a Relation has from its Digraph – EXAMPLE 2



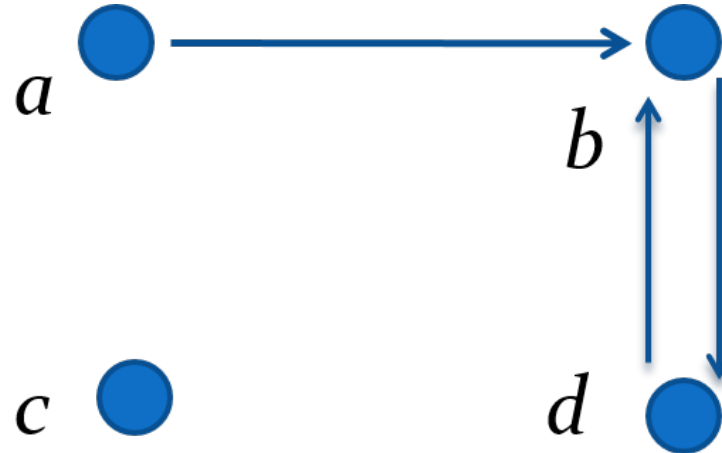
Reflexive? No, not every vertex has a loop

Symmetric? Yes, there is no edge from one vertex to another

Antisymmetric? Yes, there is no edge from one vertex to another

Transitive? Yes, since there is no edge from one vertex to another

Determining which Properties a Relation has from its Digraph – EXAMPLE 3



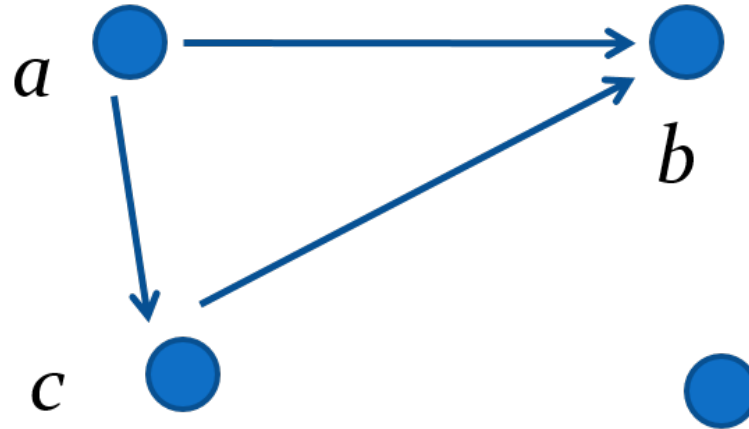
Reflexive? No, there are no loops

Symmetric? No, there is an edge from a to b , but not from b to a

Antisymmetric? No, there is an edge from d to b and b to d

Transitive? No, there are edges from a to b and from b to d , but there is no edge from a to d

Determining which Properties a Relation has from its Digraph – EXAMPLE 4



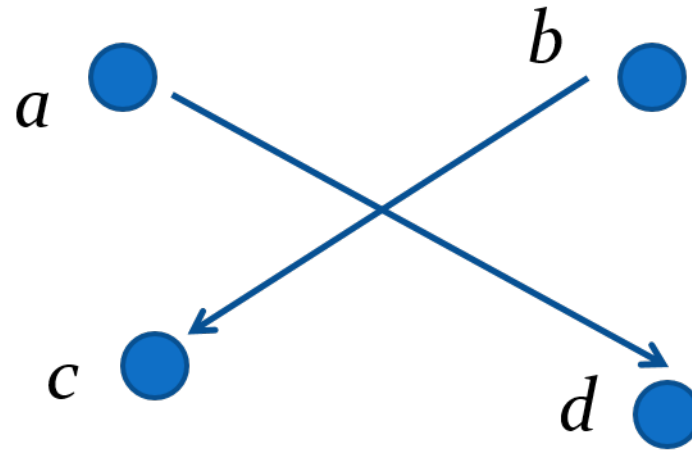
Reflexive? No, there are no loops

Symmetric? No, for EXAMPLE, there is no edge from c to a

Antisymmetric? Yes, whenever there is an edge from one vertex to another, there is not one going back

Transitive? Yes

Determining which Properties a Relation has from its Digraph – EXAMPLE 5



Reflexive? No, there are no loops

Symmetric? No, for EXAMPLE, there is no edge from d to a

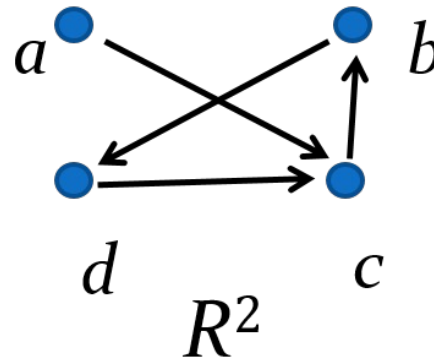
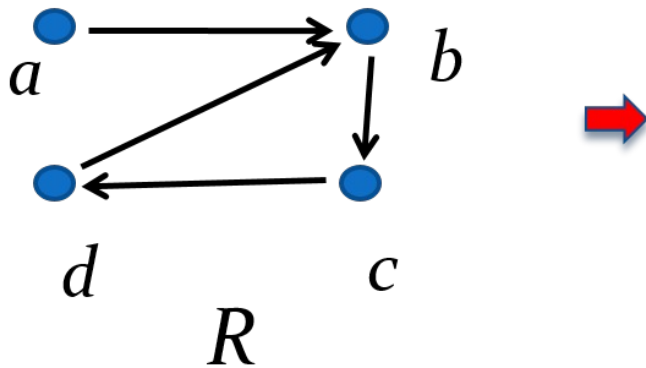
Antisymmetric? Yes, whenever there is an edge from one vertex to another, there is not one going back

Transitive? Yes, there are no two edges where the first edge ends at the vertex where the second edge begins

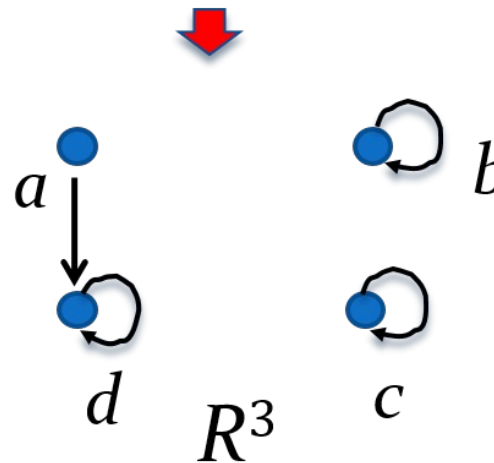
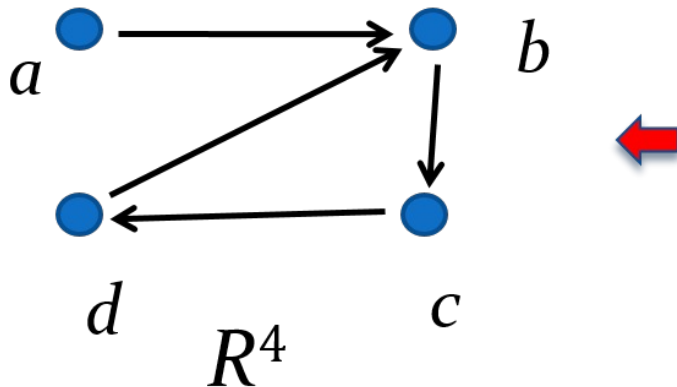
EXAMPLE of the Powers of a Relation

$$A^{[r]} = \underbrace{A \odot A \odot \dots \odot A}_{r \text{ times}}$$

$$M_{R^2} = M_R^{[2]} = M_R \odot M_R$$



The pair (x,y) is in R^n if there is a path of length n from x to y in R (following the direction of the arrows).



Section Summary

Representing Relations using **Matrices**

Representing Relations using **Digraphs**

Homework after class

9.3 Representing Relations

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Closures of Relations

Section 9.4

9.4.1 Introduction

Let R be the relation containing (a, b) if there is a telephone line from the data center in a to that in b .

How can we determine if there is some (possibly indirect) link composed of one or more telephone lines from one center to another?

Because not all links are direct, such as the link from Beijing to Shenzhen that goes through Hefei, R cannot be used directly to answer this.

In the language of relations, R is not transitive, so it does not contain all the pairs that can be linked.

Transitive Closure of Relation

As we will show in this section, we can find all pairs of data centers that have a link by constructing a **transitive relation S** containing **R** such that **S** is a subset of every transitive relation containing **R** .

Here, **S** is the **smallest transitive relation** that contains **R** .

This relation is called the **transitive closure of R** .

9.4.2 Different Types of Closures

If R is a relation on a set A , it may or may not have some property P , such as reflexivity, symmetry, or transitivity.

When R does not enjoy property P , we would like to find the smallest relation S on A with property P that contains R .

DEFINITION 1: If R is a relation on a set A , then the closure of R with respect to P , if it exists, is the relation S on A with property P that contains R and is a subset of every subset of $A \times A$ containing R with property P .

9.4.2 Different Types of Closures

If there is a **relation S that is a subset** of every relation containing R with property P , it must be **unique**.

To see this, suppose that relations S and T both have property P and are subsets of every relation with property P that contains R .

Then, S and T are subsets of each other, and so are **equal**.

Such a relation, if it exists, is the **smallest relation** with property P that contains R because it is a subset of every relation with property P that contains R .

We will show how **reflexive, symmetric, and transitive closures** of relations can be **found** (also may be not exist).

Reflexive Closure of a Relation

EXAMPLE: The relation $R = \{(1, 1), (1, 2), (2, 1), (3, 2)\}$ on the set $A = \{1, 2, 3\}$ is not reflexive. How can we produce a reflexive relation containing R that is as small as possible?

$$\{(1, 1), (1, 2), (2, 1), (3, 2), (2, 2), (3, 3)\}$$

Because this relation contains R , is reflexive, and is contained within every reflexive relation that contains R , it is called the **reflexive closure of R** .

Reflexive Closure of a Relation

Given a relation R on a set A , the reflexive closure of R can be formed by adding to R all pairs of the form (a, a) with $a \in A$, not already in R .

The addition of these pairs produces a new relation that is reflexive, contains R , and is contained within any reflexive relation containing R .

We see that the reflexive closure of R equals $R \cup \Delta$, where $\Delta = \{(a, a) \mid a \in A\}$ is the diagonal relation on A .

Reflexive Closure of a Relation

EXAMPLE: What is the reflexive closure of the relation $R = \{(a, b) \mid a < b\}$ on the set of integers?

Solutions: The reflexive closure of R is

$$\begin{aligned} R \cup \Delta &= \{(a, b) \mid a < b\} \cup \{(a, a) \mid a \in \mathbb{Z}\} \\ &= \{(a, b) \mid a \leq b\}. \end{aligned}$$

Symmetric Closure of a Relation

EXAMPLE: The relation $\{(1, 1), (1, 2), (2, 2), (2, 3), (3, 1), (3, 2)\}$ on $\{1, 2, 3\}$ is not **symmetric**. How can we produce a symmetric relation that is as small as possible and contains R ?

$\{(1, 1), (1, 2), (2, 2), (2, 3), (3, 1), (3, 2), (2, 1), (1, 3)\}$

This new relation is **symmetric** and **contains** R , it is called the **symmetric closure** of R .

Symmetric Closure of a Relation

The **symmetric closure** of a relation R can be constructed by **adding** all ordered pairs of the form (b, a) , where (a, b) is in the relation, that are not already present in R .

The symmetric closure of a relation can be constructed by taking the **union** of a relation with its **inverse**; that is, $R \cup R^{-1}$ is the symmetric closure of R , where $R^{-1} = \{(b, a) \mid (a, b) \in R\}$.

Symmetric Closure of a Relation

EXAMPLE: What is the **symmetric closure** of the relation $R = \{(a, b) \mid a > b\}$ on the set of positive integers?

Solution: The symmetric closure of R is the relation

$$\begin{aligned} R \cup R^{-1} &= \{(a, b) \mid a > b\} \cup \{(b, a) \mid a > b\} \\ &= \{(a, b) \mid a \neq b\}. \end{aligned}$$

Transitive Closure of a Relation ?

Suppose that a relation R is not transitive.

How can we produce a transitive relation that contains R such that this new relation is contained within any transitive relation that contains R ?

Can the transitive closure of a relation R be produced by adding all the pairs of the form (a, c) , where (a, b) and (b, c) are already in the relation?

Transitive Closure of a Relation ?

Consider the relation $R = \{(1, 3), (1, 4), (2, 1), (3, 2)\}$ on the set $\{1, 2, 3, 4\}$.

This relation is **not transitive**.

$\{(1, 3), (1, 4), (2, 1), (3, 2), (1, 2), (2, 3), (2, 4), \text{ and } (3, 1)\}$.

$\{(1, 3), (1, 4), (2, 1), (3, 2), (1, 2), (2, 3), (2, 4), \text{ and } (3, 1)\}$.

Adding these pairs does not produce a transitive relation, because here is **not contain** $(3, 4)$.

This shows that constructing the transitive closure of a relation is **more complicated** than constructing either the reflexive or symmetric closure.

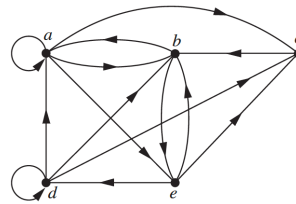
Transitive Closure of a Relation ?

The rest of this section develops algorithms for constructing transitive closures.

The transitive closure of a relation can be found by adding new ordered pairs that must be present and then repeating this process until no new ordered pairs are needed.

9.4.3 Paths in Directed Graphs

Relations by **directed graphs** helps in the construction of **transitive closures**.



DEFINITION 2:

A **path** from **a** to **b** in the directed graph **G** is a sequence of **edges** $(x_0, x_1), (x_1, x_2), (x_2, x_3), \dots, (x_{n-1}, x_n)$ in **G**, where n is a nonnegative integer, and $x_0 = a$ and $x_n = b$, that is, a sequence of edges where the terminal vertex of an edge is the same as the initial vertex in the next edge in the path.

This path is denoted by $x_0, x_1, x_2, \dots, x_{n-1}, x_n$ and has **length n**.

We view the empty set of edges as a path of **length zero** from **a** to **a**.

A path of length $n \geq 1$ that begins and ends at the same vertex is called a **circuit** or **cycle**.

9.4.3 Paths in Directed Graphs

Which of the following are **paths** in the directed graph ? What are the **lengths** of those that are paths?

Which of the paths in this list are circuits?

begins and ends at the same vertex

✓ ☐ a, b, e, d 3

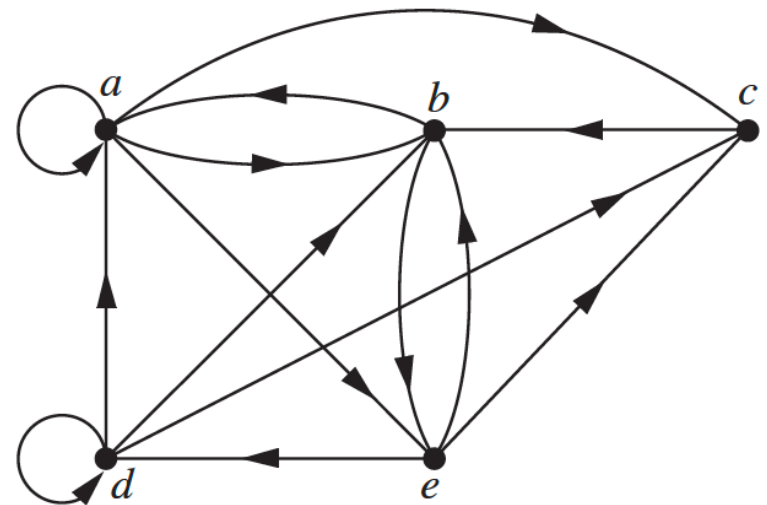
✗ ☐ a, e, c, d, b

✓ ☐ **b**, a, c, b, a, a, **b** 6

✓ ☐ d, c 1

✓ ☐ c, b, a 2

✓ ☐ **e**, b, a, b, a, b, **e** 6



Path in a Relation

The term path also applies to relations.

Carrying over the definition from directed graphs to relations, there is a **path** from **a** to **b** in **R** if there is a sequence of elements $a, x_1, x_2, \dots, x_{n-1}, b$ with $(a, x_1) \in R, (x_1, x_2) \in R, \dots$, and $(x_{n-1}, b) \in R$.

THEOREM 1: Let R be a relation on a set A . There is a path of length n , where n is a positive integer, from a to b if and only if $(a, b) \in R^n$.

Path in a Relation

THEOREM 1: Let R be a relation on a set A . There is a path of length n , where n is a positive integer, from a to b if and only if $(a, b) \in R^n$.

Proof: We will use **mathematical induction**.

$n=1$

$$(a, b) \in R$$

R^n : Powers of a Relation ($R^n = R^{n-1} \circ R$).

$A^{[r]}$: The r th Boolean power of A

Assume that the THEOREM is true for the positive integer n

$$(a, c) \in R \text{ and } (c, b) \in R^n$$

then

$$(a, b) \in R^{n+1}$$

$n+1$

Path in a Relation

THEOREM 1: Let R be a relation on a set A . There is a path of length n , where n is a positive integer, from a to b if and only if $(a, b) \in R^n$.

Proof: We will use **mathematical induction**.

By DEFINITION, there is a path from a to b of length one if and only if $(a, b) \in R$, so the THEOREM is true when $n = 1$.

Assume that the THEOREM is true for the positive integer n . This is the inductive hypothesis. There is a path of length $n + 1$ from a to b if and only if there is an element $c \in A$ such that there is a path of length one from a to c , so $(a, c) \in R$, and a path of length n from c to b , that is, $(c, b) \in R^n$.

Consequently, by the inductive hypothesis, there is a path of length $n + 1$ from a to b if and only if there is an element c with $(a, c) \in R$ and $(c, b) \in R^n$.

But there is such an element if and only if $(a, b) \in R^{n+1}$.

Therefore, there is a path of length $n + 1$ from a to b if and only if $(a, b) \in R^{n+1}$.

This completes the proof.

9.4.4 Transitive Closures

We now show that finding the **transitive closure of a relation** is **equivalent** to determining which **pairs of vertices** in the associated **directed graph** are connected by a path.

With this in mind, we define **a new relation**:

Connectivity Relation (R^*)

Connectivity Relation R^*

DEFINITION 3: Let R be a relation on a set A . The **connectivity relation R^*** consists of the pairs (a, b) such that there is a **path of length at least one** from a to b in R .

Because R^n consists of the pairs (a, b) such that there is a path of length n from a to b , it follows that R^* is the union of all the sets R^n .

$$R^* = \bigcup_{n=1}^{\infty} R^n.$$

The connectivity relation is useful in many models.

Connectivity Relation R^*

EXAMPLE: Let R be the relation on the set of all people in the world that contains (a, b) if a has met b . What is R^n , where n is a positive integer greater than one? What is R^* ?

Solution:

The relation R^2 contains (a, b) if there is a person c such that $(a, c) \in R$ and $(c, b) \in R$. Similarly, R^n consists of those pairs (a, b) such that there are people $x_1, x_2, \dots, x_{n-1}$ such that a has met x_1 , x_1 has met $x_2, \dots$, and x_{n-1} has met b .

The relation R^* contains (a, b) if there is a sequence of people, starting with a and ending with b , such that each person in the sequence has met the next person in the sequence.

Connectivity Relation R^*

EXAMPLE: Let R be the relation on the set of all subway stops in City that contains (a, b) if it is possible to travel from stop a to stop b without changing trains.

What is R^n when n is a positive integer? What is R^* ?

Solution:

The relation R^n contains (a, b) if it is possible to travel from stop a to stop b by making at most $n - 1$ changes of trains.

The relation R^* consists of the ordered pairs (a, b) where it is possible to travel from stop a to stop b making as many changes of trains as necessary.

Connectivity Relation R^*

EXAMPLE: Let R be the relation on the set of all states in the United States that contains (a, b) if state a and state b have a common border.

What is R^n , where n is a positive integer? What is R^* ?

Solution:

The relation R^n consists of the pairs (a, b) , where it is possible to go from state a to state b by crossing exactly n state borders.

R^* consists of the ordered pairs (a, b) , where it is possible to go from state a to state b crossing as many borders as necessary.

The only ordered pairs not in R^* are those containing states that are not connected to the continental United States (that is, those pairs containing Alaska or Hawaii).

Connectivity Relation R^*



Transitive Closure and Connectivity

THEOREM 2: The transitive closure of a relation R **equals** the connectivity relation R^* .

LEMMA 1:

Let A be a set with n elements, and let R be a relation on A .

If there is a **path** of length at least one in R from a to b , then there is such a path with length **not exceeding n** . (Note that there may also be more than n .)

Moreover, when $a \neq b$, if there is a path of length at least one in R from a to b , then there is such a path with length **not exceeding $n - 1$** . (Note that there may also be more than $n-1$.)

Pigeonhole Principle (Chapter 6.2): If k is a positive integer and $k + 1$ objects are placed into k boxes, then at least one box contains two or more objects.

Transitive Closure

$$R^* = \bigcup_{n=1}^{\infty} R^n.$$

$$R^* = R \cup R^2 \cup R^3 \cup \dots \cup R^n$$

THEOREM 3: Let M_R be the zero–one matrix of the relation R on a set with n elements. Then the zero–one matrix of the transitive closure R^* is

$$M_{R^*} = M_R \vee M_R^{[2]} \vee M_R^{[3]} \vee \dots \vee M_R^{[n]}. \quad M_{R^n} = M_R^{[n]}$$

This can be used as a basis for an algorithm for computing the matrix of the relation R^* .

```
procedure transitive closure ( $M_R$  : zero–one  $n \times n$  matrix)
   $A := M_R$ 
   $B := A$ 
  for  $i := 2$  to  $n$ 
     $A := A \odot M_R$ 
     $B := B \vee A$ 
  return  $B$  { $B$  is the zero–one matrix for  $R^*$ }
```


Transitive Closure

$$M_{R^*} = M_R \vee M_R^{[2]} \vee M_R^{[3]} \vee \dots \vee M_R^{[n]}.$$

EXAMPLE: Find the zero-one matrix of the **transitive closure** of the relation R where

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Solution:

$$\mathbf{M}_{R^*} = \mathbf{M}_R \vee \mathbf{M}_R^{[2]} \vee \mathbf{M}_R^{[3]}.$$

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \quad \mathbf{M}_R^{[2]} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \quad \mathbf{M}_R^{[3]} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\mathbf{M}_{R^*} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

Transitive Closure

$$M_{R^*} = M_R \vee M_R^{[2]} \vee M_R^{[3]} \vee \dots \vee M_R^{[n]}.$$

```
procedure transitive closure ( $\mathbf{M}_R$  : zero-one  $n \times n$  matrix)
   $\mathbf{A} := \mathbf{M}_R$ 
   $\mathbf{B} := \mathbf{A}$ 
  for  $i := 2$  to  $n$ 
     $\mathbf{A} := \mathbf{A} \odot \mathbf{M}_R$ 
     $\mathbf{B} := \mathbf{B} \vee \mathbf{A}$ 
  return  $\mathbf{B}$  { $\mathbf{B}$  is the zero-one matrix for  $R^*$ }
```

Computing the Boolean powers $M_R, M_R^{[2]}, M_R^{[2]}, \dots, M_R^{[n]}$ (A in code) requires that

- $n-1$ Boolean products of $n \times n$ zero-one matrices.
- Each of these Boolean products can be found using $n^2(2n-1)$ bit operations.

To find M_{R^*} (B in code) from the n Boolean powers of M_R :

- $n-1$ joins of zero-one matrices need to be found.
- Computing each of these joins uses n^2 bit operations.

Therefore, $(n-1) \cdot n^2(2n-1) + (n-1) \cdot n^2 = 2n^3(n-1)$, which is $O(n^4)$ bit operations.

We need a **more efficient algorithm** for finding transitive closures.

9.4.5 Warshall's Algorithm

The concept of the **interior vertices** of a path is used in Warshall's algorithm.

If $a, x_1, x_2, \dots, x_{m-1}, b$ is a path, its **interior vertices** are $x_1, x_2, \dots, x_{m-1}$, that is.

For instance, the interior vertices of a path a, c, d, f, g, h, b, j in a directed graph are c, d, f, g, h , and b .

The interior vertices of a, c, d, a, f, b are c, d, a , and f .

Note that the **first vertex** in the path is **not an interior vertex** unless it is visited again by the path, except as the last vertex.

Similarly, the **last vertex** in the path is **not an interior vertex** unless it was visited previously by the path, except as the first vertex.

9.4.5 Warshall's Algorithm

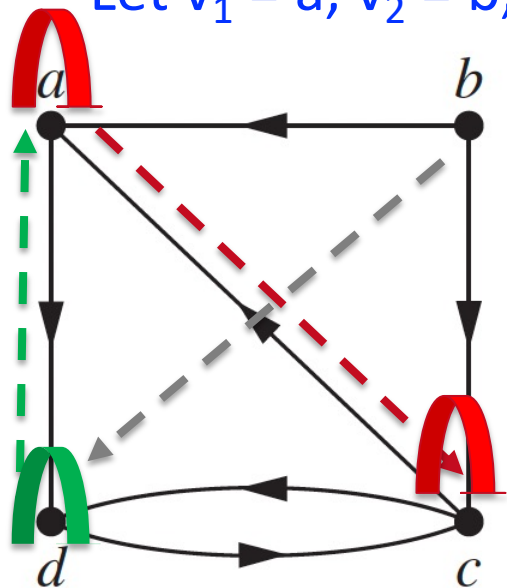
Warshall's algorithm is based on the construction of a sequence of **zero-one matrices**. These matrices are $W_0, W_1, \dots, W_n$, where $W_0 = M_R$ is the zero-one matrix of this relation, and $W_k = [W_{ij}^{(k)}]$, where $W_{ij}^{(k)} = 1$ if there is a path from v_i to v_j such that all the interior vertices of this path are in the set $\{v_1, v_2, \dots, v_k\}$ (the **first k vertices** in the list) and is 0 otherwise.

Note that $W_n = M_{R^*}$, because the (i, j) th entry of M_{R^*} is 1 if and only if there is a path from v_i to v_j , with all interior vertices in the set $\{v_1, v_2, \dots, v_n\}$ (these are the only vertices in the directed graph).

9.4.5 Warshall's Algorithm

EXAMPLE: Let R be the relation with directed graph. Let a, b, c, d be a listing of the elements of the set. Find the matrices W_0, W_1, W_2, W_3 , and W_4 . The matrix W_4 is the transitive closure of R .

Let $v_1 = a, v_2 = b, v_3 = c$, and $v_4 = d$. W_0 is the matrix of the relation.



$$W_0 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$W_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$W_3 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$$W_4 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$v_1 = a, v_2 = b$

$$W_2 = W_1.$$

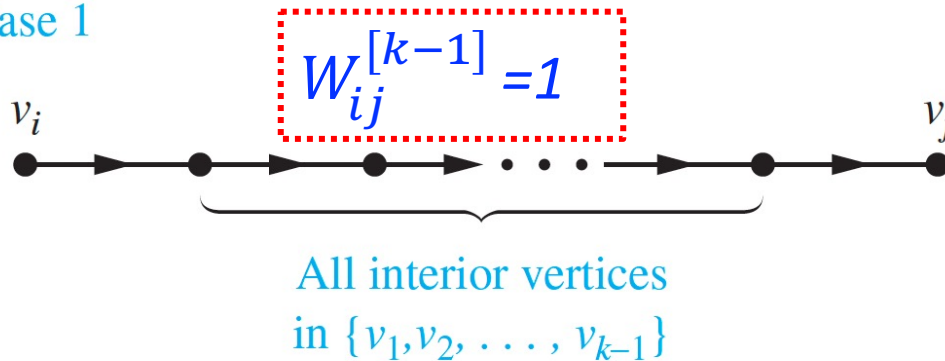
W_4 is the matrix of the transitive closure.

9.4.5 Warshall's Algorithm

Computes M_{R^*} by efficiently computing $W_1, W_2, \dots, W_n = M_{R^*}$.

We can compute W_k directly from W_{k-1} :

Case 1



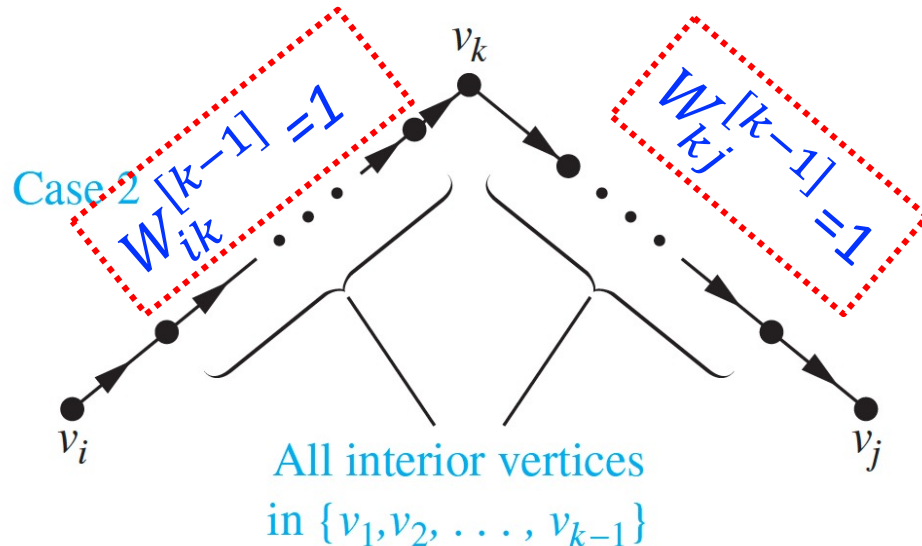
LEMMA 2

$$W_{ij}^{[k]} = W_{ij}^{[k-1]} \vee (W_{ik}^{[k-1]} \wedge W_{kj}^{[k-1]})$$

case 1
or
case 2

$$W_{ij}^{[k]} = 1$$

Case 2



9.4.5 Warshall's Algorithm

LEMMA 2 Let $W_k = [M_{ij}^{[k]}]$ be the zero-one matrix that has a 1 in its (i, j) th position if and only if there is a path from v_i to v_j with interior vertices from the set $\{v_1, v_2, \dots, v_k\}$. Then

$$W_{ij}^{[k]} = W_{ij}^{[k-1]} \vee (W_{ik}^{[k-1]} \wedge W_{kj}^{[k-1]})$$

whenever i, j , and k are positive integers not exceeding n .

```
procedure Warshall ( $\mathbf{M}_R : n \times n$  zero-one matrix)
 $\mathbf{W} := \mathbf{M}_R$ 
for  $k := 1$  to  $n$ 
    for  $i := 1$  to  $n$ 
        for  $j := 1$  to  $n$ 
             $w_{ij} := w_{ij} \vee (w_{ik} \wedge w_{kj})$ 
return  $\mathbf{W}$  {  $\mathbf{W} = [w_{ij}]$  is  $\mathbf{M}_{R^*}$  }
```

$$n \cdot 2n^2 = 2n^3$$

$$O(n^3)$$

Section Summary

Closure of a Relation

- ❑ Reflexive Closure of a Relation
- ❑ Symmetric Closure of a Relation
- ❑ Transitive Closure of a Relation

Paths in Directed Graphs

- ❑ Circuits
- ❑ Lengths

Connectivity Relation R^*

Warshall's Algorithm

Homework after class

9.4 Closures of Relations

- 1
- 3
- 26(a)

Equivalence Relations

Section 9.5

9.5.1 Introduction

In some **programming languages** the names of variables can contain an **unlimited number** of characters.

However, there is a **limit** on the number of characters that are checked when a **compiler** determines whether two variables are equal.

For instance, in traditional **C**, only the **first eight characters** of a variable name are **checked** by the compiler.

Consequently, the compiler considers strings longer than eight characters that **agree** in their first eight characters the same.

9.5.1 Introduction

Let R be the relation on the set of strings of characters such that sRt , where s and t are two strings, if s and t are at least eight characters long and the first eight characters of s and t agree, or $s = t$.

It is easy to see that R is reflexive, symmetric, and transitive.

9.5.1 Introduction

In this section we will show that such **relations** split **sets** into **disjoint classes of equivalent elements**.

Equivalence relations arise whenever we care only whether **an element** of a set is **in a certain class of elements**, instead of caring about its particular identity.

9.5.2 Equivalence Relations

DEFINITION 1: A relation on a set A is called an *equivalence relation* if it is reflexive, symmetric, and transitive.

DEFINITION 2: Two elements a , and b that are related by an equivalence relation are called *equivalent*. The notation $a \sim b$ is often used to denote that a and b are equivalent elements with respect to a particular equivalence relation.

Strings

EXAMPLE: Suppose that R is the relation on the set of strings of English letters such that aRb if and only if $l(a) = l(b)$, where $l(x)$ is the length of the string x . Is R an equivalence relation?

Solution: Show that *all of the properties* of an equivalence relation hold.

Reflexivity:

Because $l(a) = l(a)$, it follows that aRa for all strings a .

Symmetry:

Suppose that aRb . Since $l(a) = l(b)$, $l(b) = l(a)$ also holds and bRa .

Transitivity:

Suppose that aRb and bRc . Since $l(a) = l(b)$, and $l(b) = l(c)$, $l(a) = l(c)$ also holds and aRc .

Congruence Modulo m

EXAMPLE: Let m be an integer with $m > 1$. Show that the relation $R = \{(a,b) \mid a \equiv b \pmod{m}\}$ is an **equivalence relation** on the set of integers.

Reflexivity:

$a \equiv a \pmod{m}$ since $a - a = 0$ is divisible by m since $0 = 0 \cdot m$.

Symmetry:

Suppose that $a \equiv b \pmod{m}$. Then $a - b$ is divisible by m , and so $a - b = km$, where k is an integer. It follows that $b - a = (-k)m$, so $b \equiv a \pmod{m}$.

Transitivity:

Suppose that $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$.

$a - b = km$ and $b - c = lm$.

$a - c = (a - b) + (b - c) = km + lm = (k + l)m$.

Therefore, $a \equiv c \pmod{m}$.

Divides

EXAMPLE: Show that the “divides” relation on the set of positive integers is not an **equivalence relation**.

Solution: The properties of reflexivity, and transitivity do hold, but there relation is not transitive. Hence, “divides” is **not an equivalence relation**.

Reflexivity:

$a \mid a$ for all a .

Symmetry:

$2 \mid 4$, but $4 \nmid 2$. Hence, the relation is **not symmetric**.

Transitivity:

Suppose that a divides b and b divides c . Then there are positive integers k and l such that $b = ak$ and $c = bl$. Hence, $c = a(kl)$, so a divides c . Therefore, the relation is transitive.

9.5.3 Equivalence Classes

DEFINITION 3: Let R be an equivalence relation on a set A . The set of all elements that are related to an element a of A is called the *equivalence class* of a . The equivalence class of a with respect to R is denoted by $[a]_R$.

$$[a]_R = \{s \mid (a, s) \in R\}.$$

When *only one relation* is under consideration, we can write $[a]$, without the subscript R , for this equivalence class.

Example of Equivalence Classes

EXAMPLE: What is the **equivalence class** of an integer for the equivalence relation (aRb iff $a=b$ or $a=-b$)?

Solution: Because an integer is equivalent to itself and its negative in this equivalence relation,

It follows that $[a] = \{-a, a\}$.

This set contains two distinct integers unless $a = 0$.

- $[1] = \{-1, 1\}$
- $[-2] = \{-2, 2\}$
- $[0] = \{0\}$

Example: Congruence Classes Modulo 4

EXAMPLE: What are the **equivalence classes** of 0, 1, 2, and 3 for congruence modulo 4?

$$a \equiv b \pmod{4}$$

$$a \bmod 4 = b \bmod 4, \text{ or } (a-b) \bmod 4 = 0$$

$$[0] = \{\dots, -8, -4, 0, 4, 8, \dots\}.$$

$$[1] = \{\dots, -7, -3, 1, 5, 9, \dots\}.$$

$$[2] = \{\dots, -6, -2, 2, 6, 10, \dots\}.$$

$$[3] = \{\dots, -5, -1, 3, 7, 11, \dots\}.$$

Note that every integer is in **exactly one** of the four equivalence classes and that the integer **n** is in the class containing $n \bmod 4$.

Example: Congruence Classes Modulo m

$$[0]_4 = \{\dots, -8, -4, 0, 4, 8, \dots\}. \quad [1]_4 = \{\dots, -7, -3, 1, 5, 9, \dots\}.$$

$$[2]_4 = \{\dots, -6, -2, 2, 6, 10, \dots\}. \quad [3]_4 = \{\dots, -5, -1, 3, 7, 11, \dots\}.$$

This can easily be generalized, replacing 4 with any positive integer m .

The equivalence classes of the relation congruence modulo m are called the **congruence classes modulo m** .

The congruence class of an integer a modulo m is denoted by $[a]_m$, so $[a]_m = \{\dots, a - 2m, a - m, a, a + m, a + 2m, \dots\}$.

Example of Equivalence Classes

EXAMPLE: What is the equivalence class of the string 0111 with respect to the equivalence relation R_3 on the set of all bit strings? (sR_3t if and only if s and t are bit strings with $s = t$ or s and t are strings of at least three bits that start with the same three bits.)

Solution: The bit strings equivalent to 0111 are the bit strings with at least three bits that begin with 011. These are the bit strings 011, 0110, 0111, 01100, 01101, 01110, 01111, and so on.

$$[011]_{R_3} = \{011, 0110, 01100, \\ 0111, 01101, \\ 01110, \\ 01111, \dots\}.$$

$$[011]_{R_3} = \{011, 0110, 0111, 01100, 01101, 01110, 01111, \dots\}.$$

9.5.4 Equivalence Classes and Partitions

THEOREM 1: let R be an equivalence relation on a set A .

These statements for elements a and b of A are **equivalent**:

(i) aRb

(ii) $[a] = [b]$

(iii) $[a] \cap [b] \neq \emptyset$

(i) implies (ii),

(ii) implies (iii)

(iii) implies (i)

Equivalence Classes and Partitions

THEOREM 1: let R be an equivalence relation on a set A .

These statements for elements a and b of A are **equivalent**:

$$(i) \underline{aRb} \qquad (ii) [a] = [b] \qquad (iii) [a] \cap [b] \neq \emptyset$$

Proof: We first show that (i) implies (ii).

Assume that aRb . We will prove that $[a] = [b]$ by showing $[a] \subseteq [b]$ and $[b] \subseteq [a]$.

Suppose $c \in [a]$. Then aRc .

Because aRb and R is symmetric, we know that bRa .

Furthermore, because R is transitive and bRa and aRc , it follows that bRc .

Hence, $c \in [b]$. This shows that $[a] \subseteq [b]$.

The proof that $[b] \subseteq [a]$ is similar;

Since $[a] \subseteq [b]$ and $[b] \subseteq [a]$, we have shown that $[a] = [b]$.

Equivalence Classes and Partitions

THEOREM 1: let R be an equivalence relation on a set A .

These statements for elements a and b of A are **equivalent**:

(i) aRb

(ii) $[a] = [b]$

(iii) $[a] \cap [b] \neq \emptyset$

Proof: we will show that (ii) implies (iii).

Assume that $[a] = [b]$.

It follows that $[a] \cap [b] \neq \emptyset$ because $[a]$ is nonempty (because $a \in [a]$ because R is reflexive).

Equivalence Classes and Partitions

THEOREM 1: let R be an equivalence relation on a set A .

These statements for elements a and b of A are **equivalent**:

(i) aRb

(ii) $[a] = [b]$

(iii) $[a] \cap [b] \neq \emptyset$

Proof: we will show that (iii) implies (i).

Suppose that $[a] \cap [b] \neq \emptyset$.

Then there is an element c with $c \in [a]$ and $c \in [b]$.

In other words, aRc and bRc . By the symmetric property, cRb .

Then by transitivity, because aRc and cRb , we have aRb .

Equivalence Classes and Partitions

THEOREM 1: let R be an equivalence relation on a set A .

These statements for elements a and b of A are **equivalent**:

(i) aRb

(ii) $[a] = [b]$

(iii) $[a] \cap [b] \neq \emptyset$

(i) implies (ii),

(ii) implies (iii)

(iii) implies (i)

So, the three statements, (i), (ii), and (iii), are equivalent.

Equivalence Relation Partitions a Set

Let R be an equivalence relation on a set A . The **union** of the equivalence classes of R **is all of A** , because an element a of A is in its own equivalence class $[a]_R$.

$$\bigcup_{a \in A} [a]_R = A.$$

In addition, from Theorem 1 (Pre-slide), it follows that these equivalence classes are either **equal** or **disjoint**,

so $[a]_R \cap [b]_R = \emptyset$, when $[a]_R \neq [b]_R$.

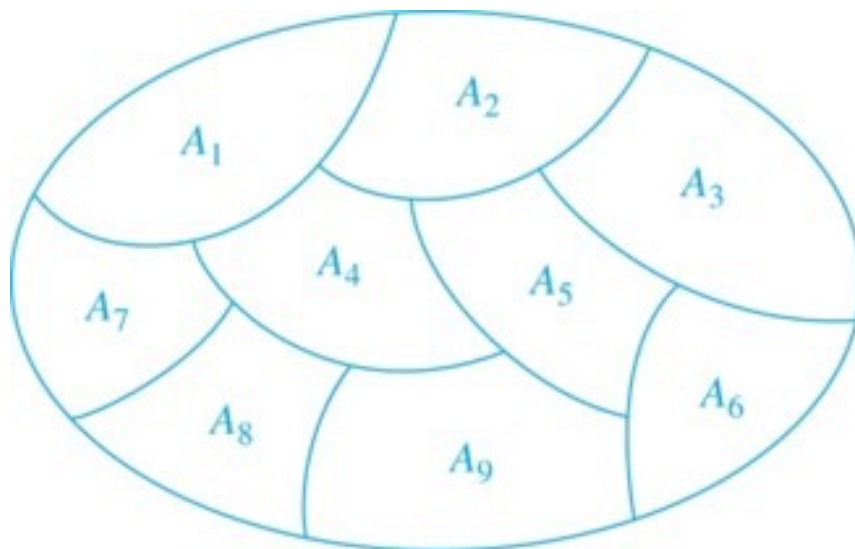
Therefore, the equivalence classes **form** a partition of A , because they split A into disjoint subsets.

Equivalence Relation Partitions a Set

More precisely, a *partition* of a set S is a collection of disjoint nonempty subsets of S that have S as their union.

In other words, the collection of subsets A_i , where $i \in I$ (where I is an index set), forms a partition of S if and only if

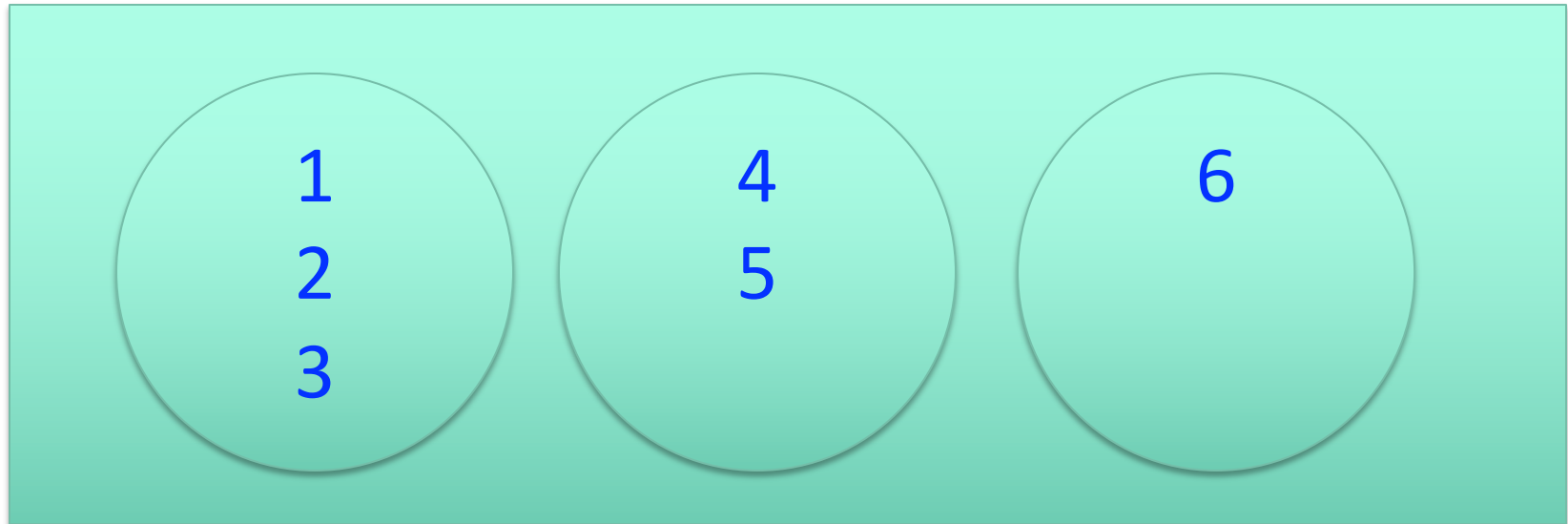
- $A_i \neq \emptyset$ for $i \in I$,
- $A_i \cap A_j = \emptyset$ when $i \neq j$,
- and $\bigcup_{i \in I} A_i = S$.



Example of Partitions a Set

EXAMPLE: Suppose that $S = \{1, 2, 3, 4, 5, 6\}$.

The collection of sets $A_1 = \{1, 2, 3\}$, $A_2 = \{4, 5\}$, and $A_3 = \{6\}$ forms a partition of S , because these sets are disjoint and their union is S .



Equivalence Relation Partitions a Set

$$S = \{1, 2, 3, 4, 5, 6\}$$

The subsets of S in this partition are equivalence classes.

$$A_1 = \{1, 2, 3\}$$

$$A_2 = \{4, 5\}$$

$$A_3 = \{6\}$$



the equivalence classes of an
equivalence relation on a set

partition of the set

Two elements are equivalent with respect to this relation if and only if they are **in the same subset of S in the partition.**

Equivalence Relation Partitions a Set

Theorem 2:

Let R be an equivalence relation on a set S . Then the equivalence classes of R form a **partition** of S .

Conversely, given a partition $\{A_i \mid i \in I\}$ of the set S , there is an equivalence relation R that has the sets $A_i, i \in I$, as its **equivalence classes**.



Proof ?

Reflexivity
Symmetry
Transitivity

Equivalence Relation Partitions a Set

Theorem 2 (second part): Given a partition $\{A_i \mid i \in I\}$ of the set S , there is an equivalence relation R that has the sets A_i , $i \in I$, as its equivalence classes.

Proof:

Reflexivity: For every $a \in S$, $(a, a) \in R$, because a is in the same subset as itself.

Symmetry: If $(a, b) \in R$, then b and a are in the same subset of the partition, so $(b, a) \in R$.

Transitivity: If $(a, b) \in R$ and $(b, c) \in R$, then a and b are in the same subset of the partition, as are b and c .
Since the subsets are disjoint and b belongs to both, the two subsets of the partition must be identical.
Therefore, $(a, c) \in R$ since a and c belong to the same subset of the partition.

Example of Partitions a Set

EXAMPLE: List the **ordered pairs** in the equivalence relation R produced by the partition $A_1 = \{1, 2, 3\}$, $A_2 = \{4, 5\}$, and $A_3 = \{6\}$ of $S = \{1, 2, 3, 4, 5, 6\}$.

Solution: The subsets of S in the partition are the equivalence classes of R . The pair $(a, b) \in R$ if and only if a and b are in the same subset of the S in the partition.

{

$(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3) ,$

$(4, 4), (4, 5), (5, 4), (5, 5) ,$

$(6, 6)$

}

Equivalence Relation Partitions a Set

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A_1 = \{1, 2, 3\}$$

$$A_2 = \{4, 5\}$$

$$A_3 = \{6\}$$



- $(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)$
- $(4, 4), (4, 5), (5, 4), (5, 5)$
- $(6, 6)$

Example of Partitions a Set

EXAMPLE: What are the sets in the partition of the integers arising from congruence modulo 4?

$$[0]_4 = \{\dots, -8, -4, 0, 4, 8, \dots\}.$$

$$[1]_4 = \{\dots, -7, -3, 1, 5, 9, \dots\}.$$

$$[2]_4 = \{\dots, -6, -2, 2, 6, 10, \dots\}.$$

$$[3]_4 = \{\dots, -5, -1, 3, 7, 11, \dots\}.$$

These congruence classes are **disjoint**, and every integer is in **exactly one** of them.

These **congruence classes** form a **partition**.

Example of Partitions a Set

EXAMPLE: What are the sets in the partition of the set of all bit strings arising from the relation R_3 on the set of all bit strings? (Recall that sR_3t , where s and t are bit strings, if $s = t$ or s and t are bit strings with **at least three bits** that agree in their first three bits.)

$$[000]R_3 = \{ \quad \quad \quad \},$$

$$[001]R_3 = \{ \quad \quad \quad \},$$

$$[010]R_3 = \{ \quad \quad \quad \},$$

$$[011]R_3 = \{ \quad \quad \quad \},$$

$$[100]R_3 = \{ \quad \quad \quad \},$$

$$[101]R_3 = \{ \quad \quad \quad \},$$

$$[110]R_3 = \{ \quad \quad \quad \},$$

$$[111]R_3 = \{ \quad \quad \quad \}.$$

Example of Partitions a Set

EXAMPLE: What are the sets in the partition of the set of all bit strings arising from the relation R_3 on the set of all bit strings? (Recall that sR_3t , where s and t are bit strings, if $s = t$ or s and t are bit strings with **at least three bits** that agree in their first three bits.)

$$[000]R_3 = \{000, 0000, 0001, 00000, 00001, 00010, 00011, \dots\},$$

$$[001]R_3 = \{001, 0010, 0011, 00100, 00101, 00110, 00111, \dots\},$$

$$[010]R_3 = \{010, 0100, 0101, 01000, 01001, 01010, 01011, \dots\},$$

$$[011]R_3 = \{011, 0110, 0111, 01100, 01101, 01110, 01111, \dots\},$$

$$[100]R_3 = \{100, 1000, 1001, 10000, 10001, 10010, 10011, \dots\},$$

$$[101]R_3 = \{101, 1010, 1011, 10100, 10101, 10110, 10111, \dots\},$$

$$[110]R_3 = \{110, 1100, 1101, 11000, 11001, 11010, 11011, \dots\},$$

$$[111]R_3 = \{111, 1110, 1111, 11100, 11101, 11110, 11111, \dots\}.$$

Section Summary

Equivalence Relations

Equivalence Classes

Equivalence Classes and Partitions

Homework after class

9.5 Equivalence Relations

- 1
- 41

Partial Orderings

Section 9.6

9.6.1 Introduction

We often use **relations to order** some or all of the elements of sets.

- ❑ we order words using the relation containing pairs of words (x, y) , where **x comes before y** in the dictionary.
- ❑ We schedule projects using the relation consisting of pairs (x, y) , where x and y are tasks in a project such that **x must be completed before y begins**.
- ❑ We order the set of integers using the relation containing the pairs (x, y) , where **x is less than y** .

When we add all of the pairs of the form (x, y) to these relations, we obtain a relation that is **reflexive, antisymmetric, and transitive**.

Partial Ordering

DEFINITION 1:

A relation R on a set S is called a *partial ordering*, or *partial order*, if it is **reflexive**, **antisymmetric**, and **transitive**.

A set together with a partial ordering R is called a *partially ordered set*, or *poset*, and is denoted by (S, R) .

Members of S are called *elements of the poset*.

Partial Ordering

EXAMPLE: Show that the “greater than or equal” relation ($\geq$) is a **partial ordering** on the set of integers.

- *Reflexivity*: $a \geq a$ for every integer a .
- *Antisymmetry*: If $a \geq b$ and $b \geq a$, then $a = b$.
- *Transitivity*: If $a \geq b$ and $b \geq c$, then $a \geq c$.

It follows that $\geq$ is a partial ordering on the set of integers and $(\mathbb{Z}, \geq)$ is a poset.

Recall: a *partially ordered set*, is denoted by (S, R) .

Partial Ordering

EXAMPLE: Show that the divisibility relation ($|$) is a **partial ordering** on the set of positive integers.

- *Reflexivity:* $a | a$ for all integers a
- *Antisymmetry:* If $a | b$ and $b | a$, then $a = b$.
- *Transitivity:* Suppose that a divides b and b divides c . Then $b = ak$ and $c = bl$. Hence, $c = a(kl)$, so a divides c

$(\mathbb{Z}^+, |)$ is a poset.

Partial Ordering

EXAMPLE: Show that the inclusion relation ($\subseteq$) is a **partial ordering** on the **power set of a set S** .

- *Reflexivity:* $A \subseteq A$ whenever A is a subset of S .
- *Antisymmetry:* $A \subseteq B$ and $B \subseteq A$, then $A = B$.
- *Transitivity:* If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

$(\mathcal{P}(S), \subseteq)$ is a poset.

Not a Partial Ordering

EXAMPLE: Let R be the relation on the set of people such that xRy if x and y are people and x is older than y . Show that R is not a partial ordering.

- *Reflexivity:* $\neg xRx$, no person is older than himself or herself.
- *Antisymmetry:* xRy , then $\neg yRx$
- *Transitivity:* if xRy and yRz , then xRz

It follows that R is not a partial ordering.

Comparability and Incomparable

DEFINITION 2:

The elements a and b of a poset $(S, \preceq)$ are *comparable* if either $a \preceq b$ **or** $b \preceq a$.

When a and b are elements of S so that **neither** $a \preceq b$ **nor** $b \preceq a$, then a and b are called *incomparable*.

The symbol $\preceq$ is used to denote the relation in any poset.

Comparability and Incomparable

EXAMPLE: In the poset $(\mathbb{Z}^+, |)$, are the integers 3 and 9 comparable? Are 5 and 7 comparable?

Solution:

The integers 3 and 9 are comparable, because $3 \mid 9$.

The integers 5 and 7 are incomparable, because $5 \nmid 7$ and $7 \nmid 5$.

Totally Ordered Set

The adjective “partial” is used to describe partial orderings because pairs of elements may be incomparable.

When every two elements in the set are comparable, the relation is called a **total ordering**.

DEFINITION 3: If $(S, \preceq)$ is a poset and every two elements of S are comparable, S is called a **totally ordered** or **linearly ordered set**, and $\preceq$ is called a **total order** or a **linear order**. A totally ordered set is also called a **chain**.

Totally Ordered Set

EXAMPLE: The poset $(\mathbb{Z}, \leq)$ is totally ordered?

Yes, because $a \leq b$ or $b \leq a$ whenever a and b are integers.

EXAMPLE: The poset $(\mathbb{Z}^+, |)$ is totally ordered?

No, because it contains elements that are incomparable, such as 5 and 7.

Well-ordered Set

DEFINITION 4: $(S, \preceq)$ is a **well-ordered set** if it is a poset such that $\preceq$ is a total ordering and every nonempty subset of S **has a least element**.

EXAMPLE:

The set of ordered pairs of positive integers, $\mathbb{Z}^+ \times \mathbb{Z}^+$ (e.g. $(0,0)$), with $(a_1, a_2) \preceq (b_1, b_2)$ if $a_1 < b_1$, or if $a_1 = b_1$ and $a_2 \leq b_2$ (the lexicographic ordering), is a well-ordered set.

The set $\mathbb{Z}$, with the usual $\leq$ ordering, is not well-ordered because the set of **negative integers**, which is a subset of $\mathbb{Z}$, has **no least element**.

9.6.2 Lexicographic Order

The words in a dictionary are listed in **alphabetic**, or **lexicographic**, order, which is based on the ordering of the letters in the alphabet.

This is a special case of **an ordering of strings** on a set constructed **from** a **partial ordering on the set**.

We will show how this construction works in any poset.

Lexicographic Order

To construct a partial ordering on the Cartesian product of two posets, $(A_1, \preceq_1)$ and $(A_2, \preceq_2)$.

The lexicographic ordering $\preceq$ on $A_1 \times A_2$ is defined by specifying that one pair is less than a second pair

- if the first entry of the first pair is less than (in A_1) the first entry of the second pair,
- or if the first entries are equal, but the second entry of this pair is less than (in A_2) the second entry of the second pair.

Lexicographic Order

In other words, (a_1, a_2) is less than (b_1, b_2) , that is,

$$(a_1, a_2) < (b_1, b_2),$$

either if $a_1 <_1 b_1$ or if both $a_1 = b_1$ and $a_2 <_2 b_2$.

We obtain a partial ordering by adding equality ($=$) to the ordering $<$ on $A_1 \times A_2$.

Lexicographic Order

EXAMPLE: Determine whether $(3, 5) < (4, 8)$, whether $(5, 8) < (4, 5)$, and whether $(4, 9) < (4, 11)$ in the poset $(\mathbb{Z} \times \mathbb{Z}, \preceq)$, where $\preceq$ is the lexicographic ordering constructed from the usual $\leq$ relation on $\mathbb{Z}$.

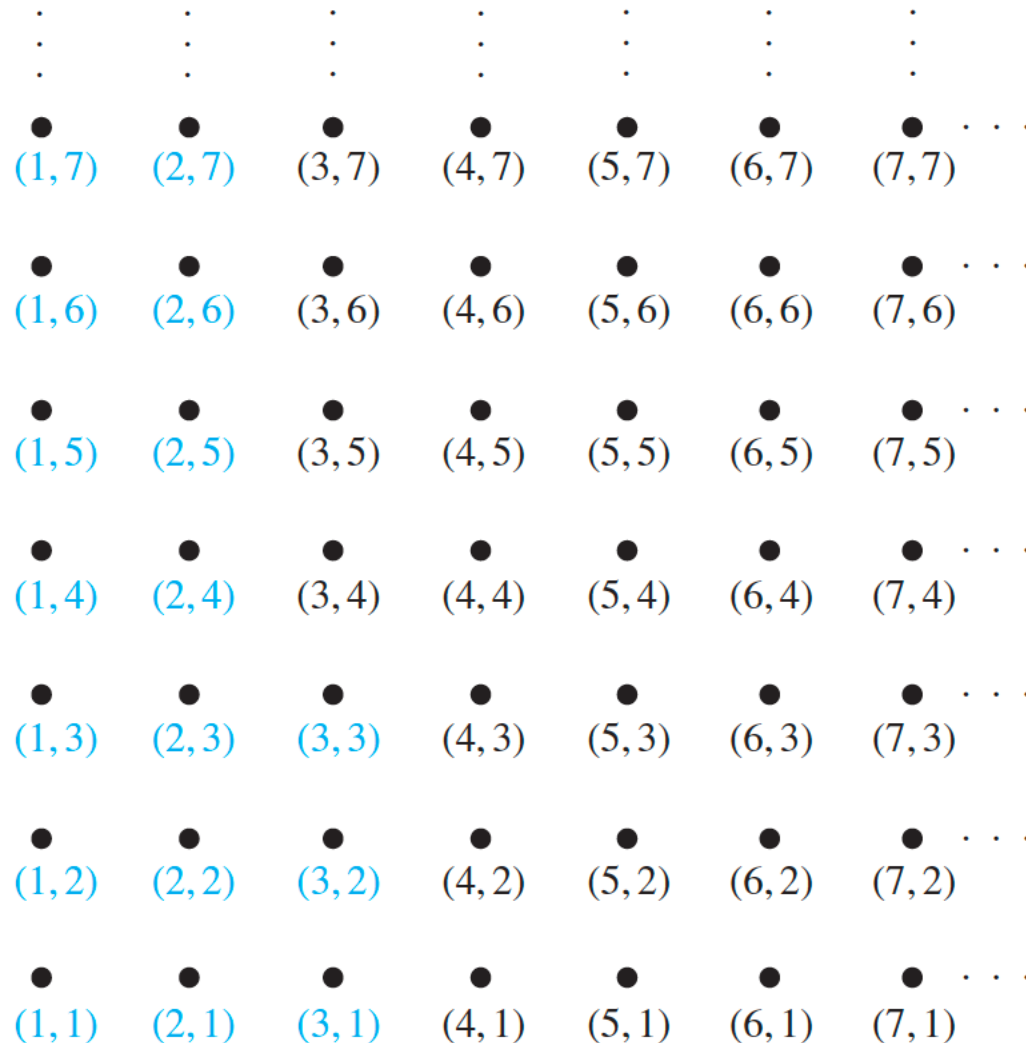
$$(3, 5) < (4, 8) \text{ Yes}$$

$$(5, 8) < (4, 5) \text{ No}$$

$$(4, 9) < (4, 11) \text{ Yes}$$

Lexicographic Order

The ordered pairs in $\mathbb{Z}^+ \times \mathbb{Z}^+$ that are less than $(3, 4)$ in lexicographic order are highlighted.



Lexicographic Order

A lexicographic ordering can be defined on the Cartesian product of n posets $(A_1, \preceq_1), (A_2, \preceq_2), \dots, (A_n, \preceq_n)$.

Define the partial ordering on $A_1 \times A_2 \times \dots \times A_n$ by $(a_1, a_2, \dots, a_n) < (b_1, b_2, \dots, b_n)$ if $a_1 <_1 b_1$, or if there is an integer $i > 0$ such that $a_1 = b_1, \dots, a_i = b_i$, and $a_{i+1} <_{i+1} b_{i+1}$.

In other words, one n -tuple is less than a second n -tuple if the entry of the first n -tuple in the first position where the two n -tuples disagree is less than the entry in that position in the second n -tuple.

EXAMPLE:

$$(1, 2, 3, 5) < (1, 2, 4, 3)$$

Lexicographic Order

$$(a_1, a_2, \dots, a_n) < (b_1, b_2, \dots, b_n)$$

This definition can be easily extended to a lexicographic ordering on strings.

EXAMPLE: Consider strings of lowercase English letters. A lexicographic ordering can be defined using the ordering of the letters in the alphabet. This is the same ordering as that used in dictionaries.

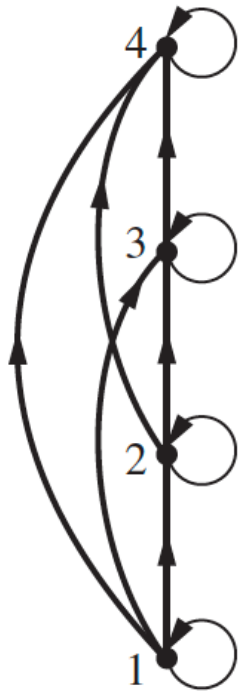
- *discreet* < *discrete*, because these strings differ in the seventh position and *e* < *t*.
- *discreet* < *discreetness*, because the first eight letters agree, but the second string is longer.

9.6.3 Hasse Diagrams

Many **edges** in the directed graph for a finite poset **do not have to be shown** because they **must be present**.

partial ordering $\{(a, b) \mid a \leq b\}$ on the set $\{1, 2, 3, 4\}$,

Because this relation is a partial ordering, it is **reflexive**, and its directed graph has **loops** at all vertices.



Because a partial ordering is **transitive**



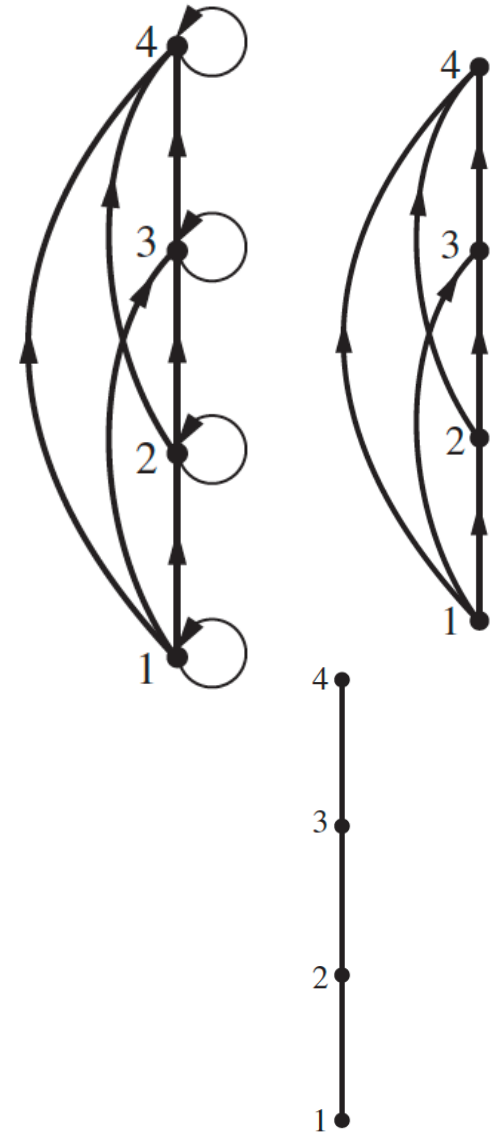
If we assume that all edges are pointed “**upward**”, we do not have to show the **directions** of the edges;

Constructing the Hasse diagram for $(\{1, 2, 3, 4\}, \leq)$.

Procedure for Constructing a Hasse Diagram

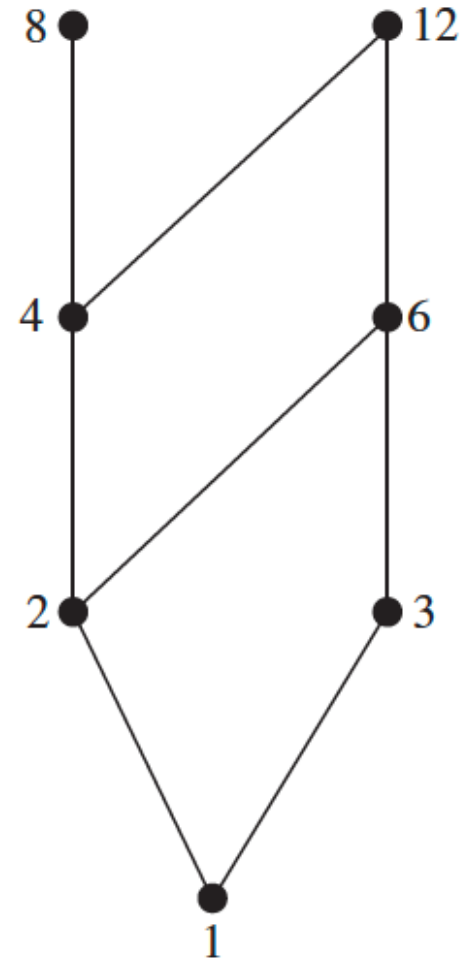
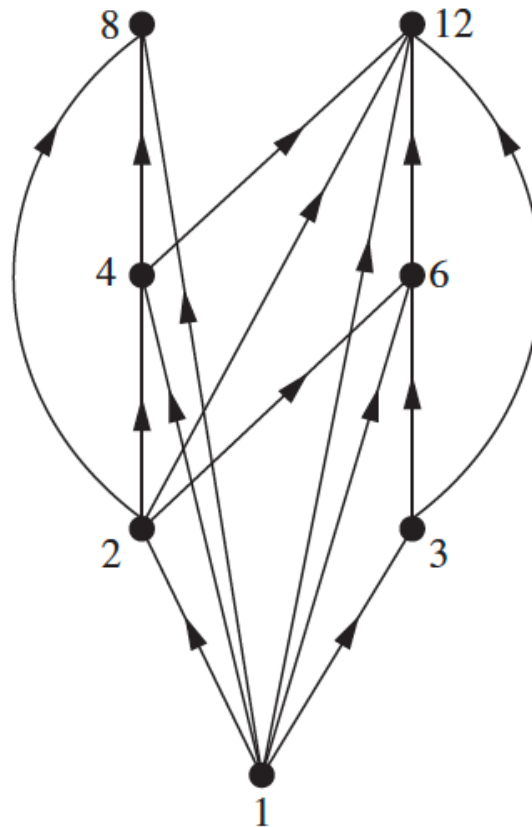
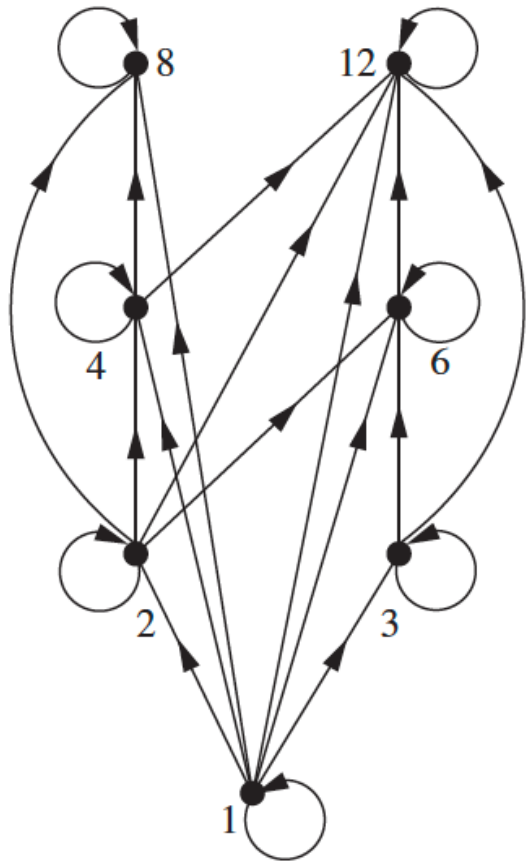
In general, we can represent a finite poset $(S, \leq)$ using this procedure:

- ❑ **Remove** the loops (a, a) present at every vertex due to the **reflexive** property.
- ❑ **Remove** all edges (x, y) for which there is an element $z \in S$ such that $x < z$ and $z < y$. These are the edges that must be present due to the **transitive** property.
- ❑ **Arrange** each edge so that its initial vertex is **below** the terminal vertex.
- ❑ **Remove** all the **arrows**, because all edges point upwards toward their terminal vertex.



Procedure for Constructing a Hasse Diagram

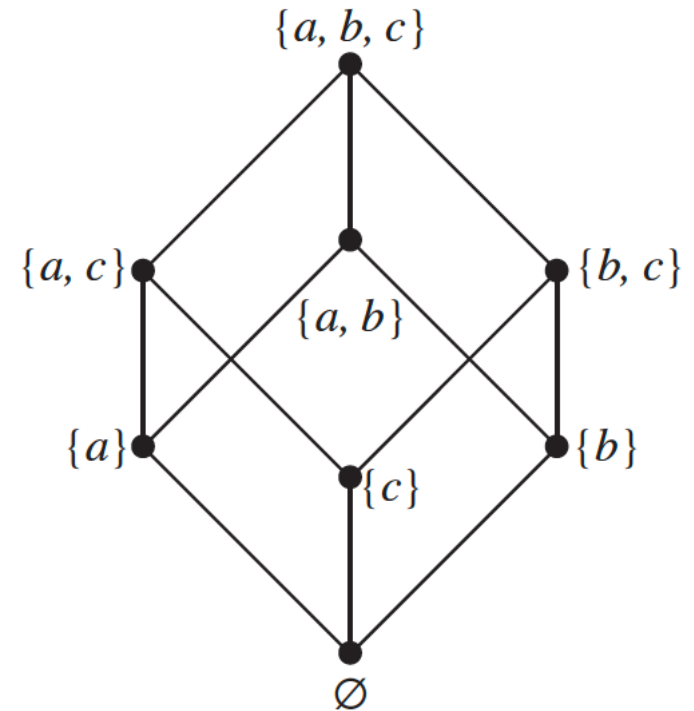
Draw the Hasse diagram representing the partial ordering $\{(a, b) \mid a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$.



Procedure for Constructing a Hasse Diagram

Draw the Hasse diagram for the partial ordering $\{(A, B) \mid A \subseteq B\}$ on the power set $\mathcal{P}(S)$, where $S = \{a, b, c\}$.

- ❑ Remove all **loops**: (a, a) , (b, b) , (c, c) ,
- ❑ Delete all the edges implied by the **transitive** property: $(\emptyset, \{a, b\})$, $(\emptyset, \{a, c\})$, $(\emptyset, \{b, c\})$, $(\emptyset, \{a, b, c\})$, $(\{a\}, \{a, b, c\})$, $(\{b\}, \{a, b, c\})$, and $(\{c\}, \{a, b, c\})$.



9.6.4 Maximal and Minimal Elements

Elements of posets that have certain **extremal** properties are important for many applications.

An element of a poset is called **maximal** if it is not less than any element of the poset.

□ a is maximal in the poset $(S, \preceq)$ if there is **no** $b \in S$ such that $a \prec b$.

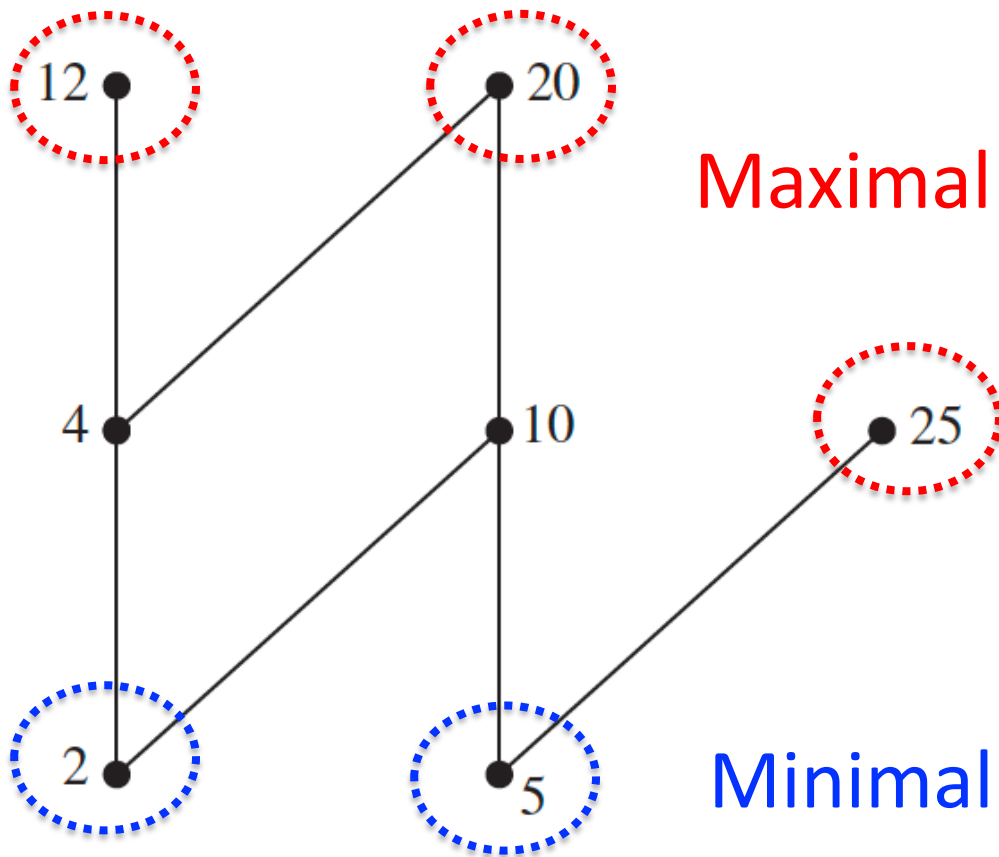
An element of a poset is called **minimal** if it is not greater than any element of the poset.

□ a is minimal if there is **no** $b \in S$ such that $b \prec a$.

Maximal and minimal elements are easy to spot using a Hasse diagram. They are the “top” and “bottom” elements in the diagram.

Maximal and Minimal Elements

EXAMPLE: Which elements of the poset $(\{2, 4, 5, 10, 12, 20, 25\}, |)$ are maximal, and which are minimal?



Maximal elements

A poset can have more than one maximal element and more than one minimal element.

Minimal elements

The Hasse diagram of a poset

Greatest and Least Elements

Sometimes there is an element in a poset that is greater than every other element.

Such an element is called the **greatest element**.

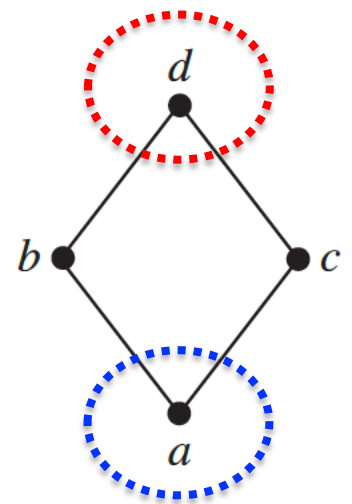
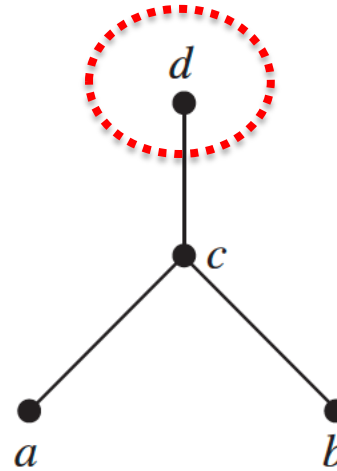
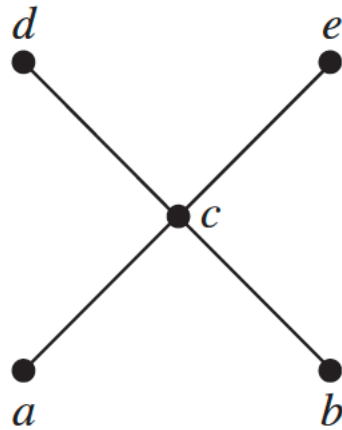
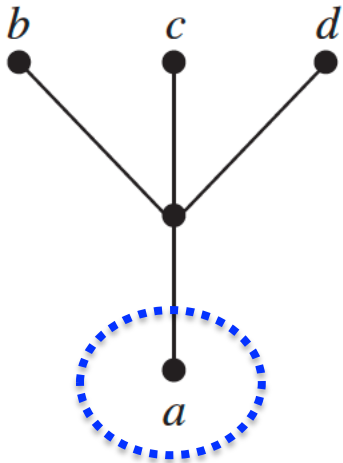
- a is the greatest element of the poset $(S, \preceq)$ if $b \preceq a$ for all $b \in S$.
- The greatest element is **unique** when it exists.

Likewise, an element is called the **least element** if it is less than all the other elements in the poset.

- a is the least element of $(S, \preceq)$ if $a \preceq b$ for all $b \in S$.
- The least element is **unique** when it exists.

Greatest and Least Elements

EXAMPLE: Determine whether the posets represented by each of the Hasse diagrams have a greatest element and a least element.



Greatest and Least Elements

EXAMPLE: Let S be a set. Determine whether there is a greatest element and a least element in the poset $(\mathcal{P}(S), \subseteq)$.

- ❑ Least element: **Empty set**, because $\emptyset \subseteq T$ for any subset T of S .
- ❑ Greatest element: **S** , because $T \subseteq S$ whenever T is a subset of S .

EXAMPLE: Is there a greatest element and a least element in the poset $(\mathbb{Z}^+, |)$?

- ❑ Least element: **1**, because $1 | n$ whenever n is a positive integer
- ❑ Greatest element: **No**, there is no integer that is divisible by all positive integers,